Welcome to
Total Quality Management
(19MEE347)

Unit 2

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CUSTOMER SATISFACTION

"An organization's success depends on how many customers it has, how much they buy, and how often they buy".

"Manufacturing and service organizations are using customer satisfaction as the measure of quality."





Introduction



- Customer satisfaction is a part of qualitative pillar of TQM. It is dynamic in nature and is subjective concept because it varies from person to person.
- Customer satisfaction has become one of the most integral or essential part of any business. In today's market, companies look at it as their business strategies or as their business requirement.

Customer satisfaction

Who is the customer?

One who uses the product or service, the one who purchases the product or service or the one who influences the product or service.

Two types of customers:

- **Internal customers**

  The customers inside the company are called internal customers.

  (eg). Every department persons

- **External customers**

  The customers outside the company are called external customers.

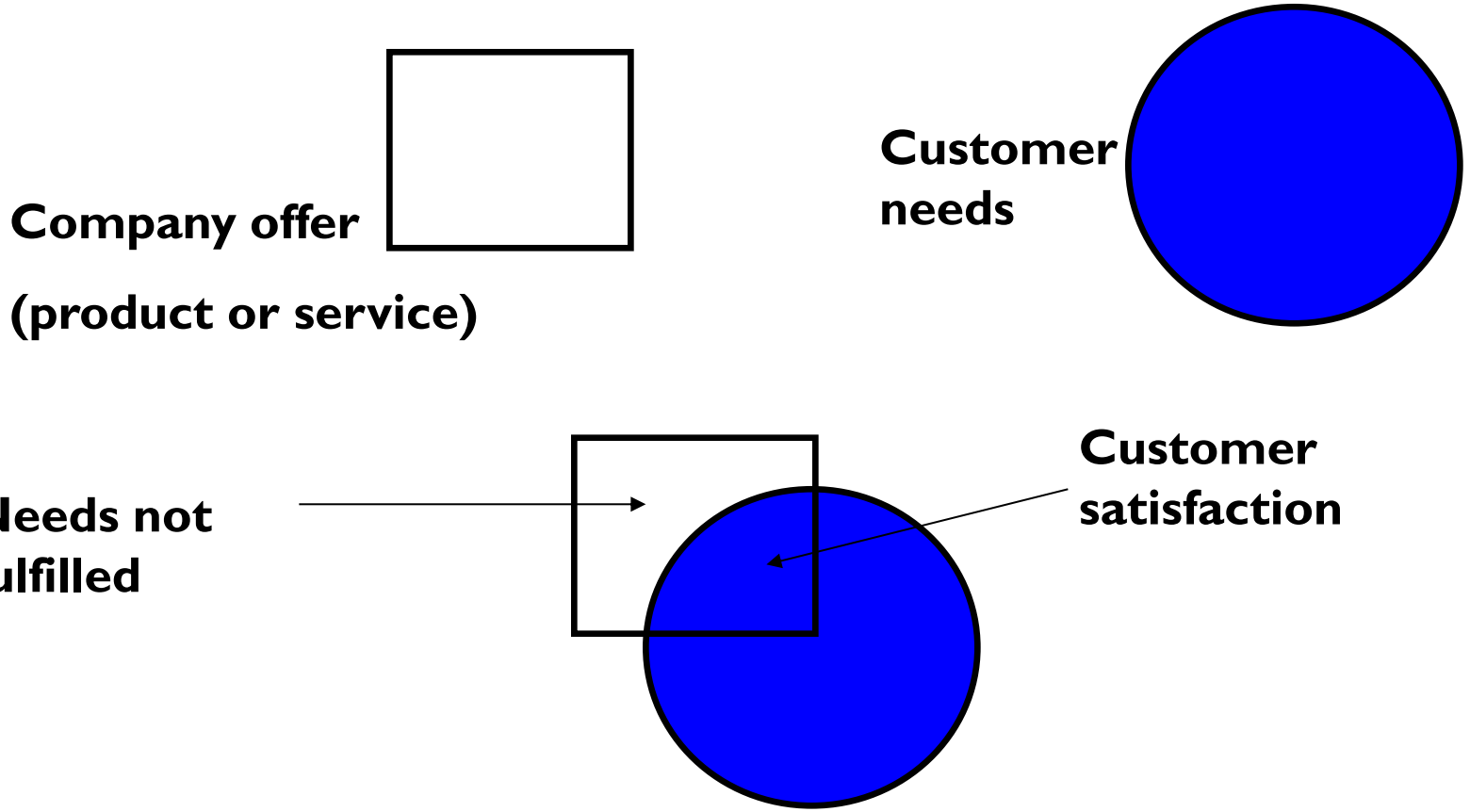
  (eg). End user, Purchaser, Merchants, Suppliers

  TQM is commitment to **customer-focus** - internal and external customers.

https://youtu.be/XK3cNcuvuMs

Customer satisfaction model

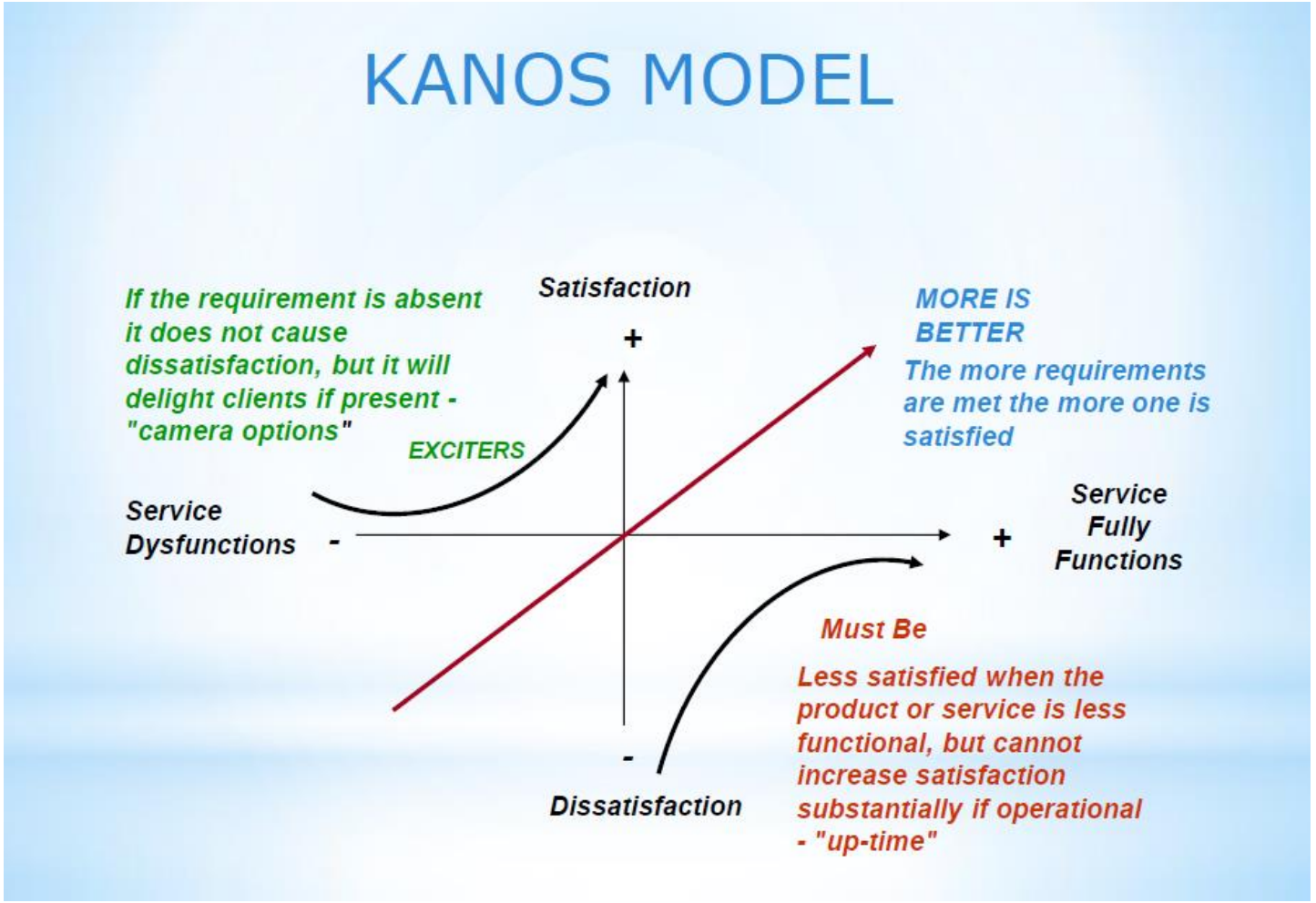
Teboul Model



Total satisfaction is achieved when offer matches the need i.e. circle is superimposed on the square.

Customer satisfaction model

Kano's Model



Understand, Integrate, and Priorities the things to attract the customers:

Improve performances

Differentiate from competitors

Ensure Success.

Bus Service

Must be: Clean seats, blanket (for AC)

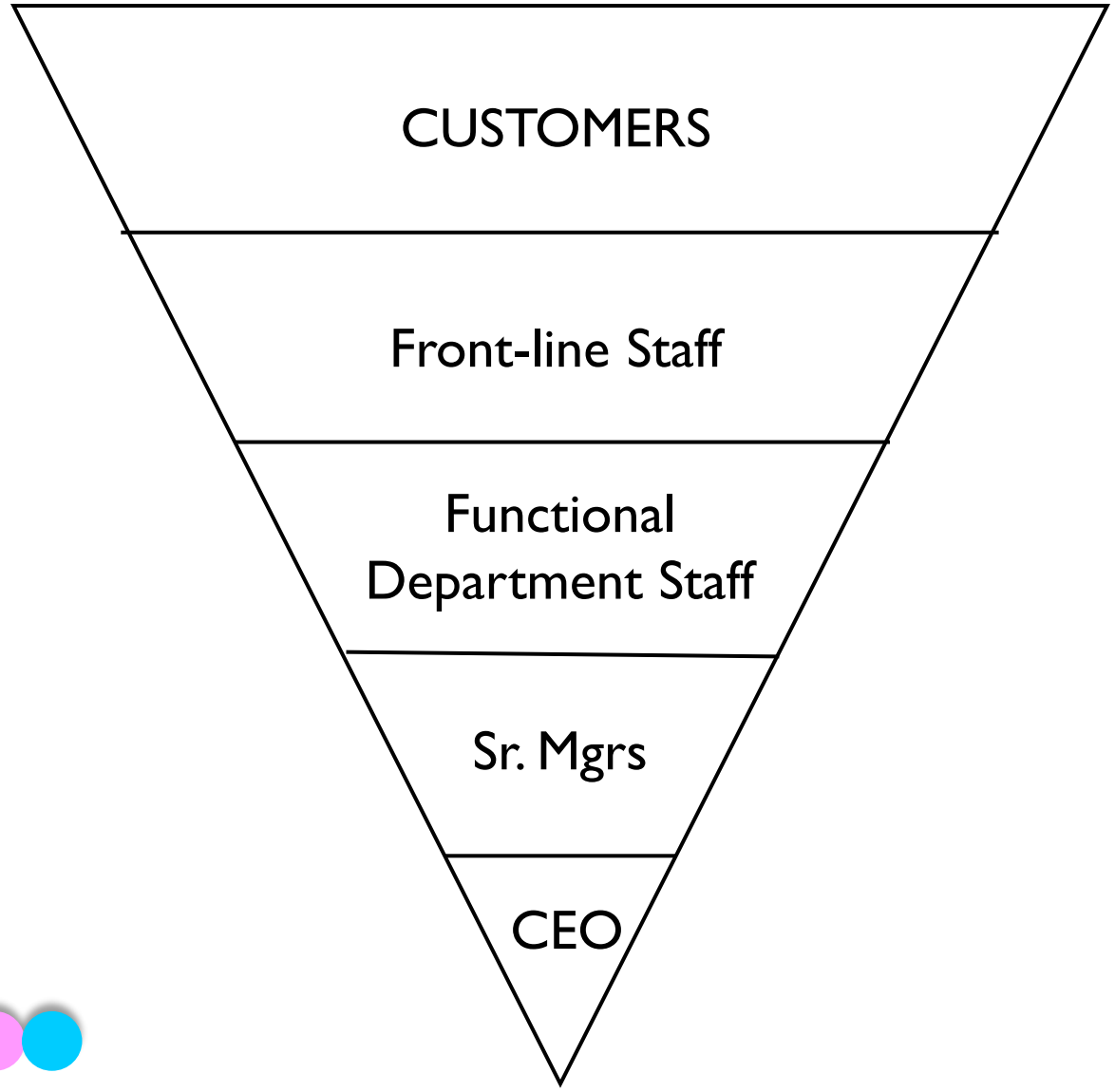
More is better: Free wifie, surprise like birthday

Exciters: Free Netflix/A-prime, drink

Customer satisfaction

- Customer is the Boss or 'King'
- Customer dictates the market trends and direction
- Customer not only has needs to be supplied (basic performance functions)
- Also he wants what he needs (additional features satisfy him and influence his purchase decision)
- Hence the Suppliers and Manufacturers have to closely follow at the heel of the customer.

Customer satisfaction organizational diagram



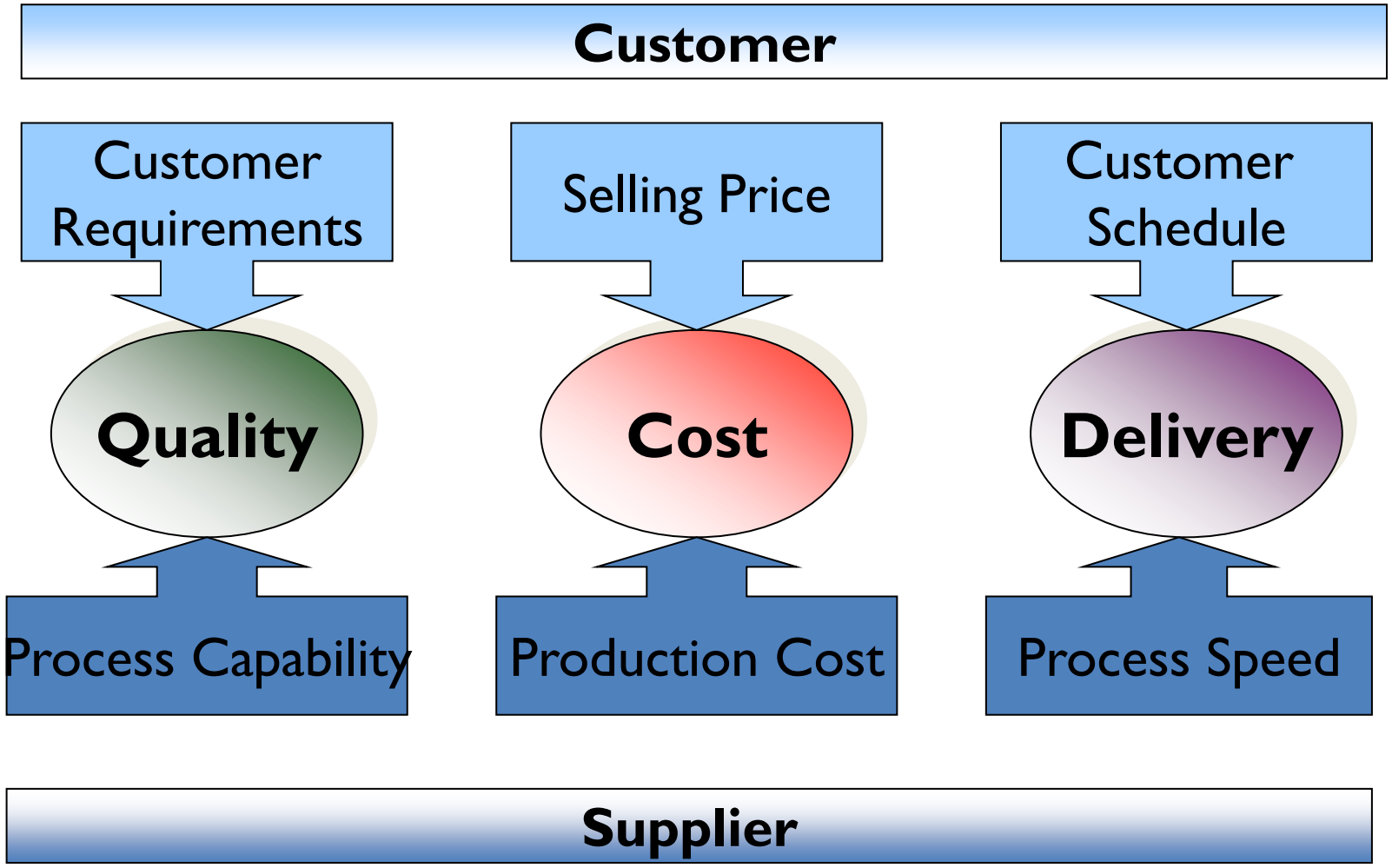
What is Customer satisfaction?

- Is it due to Product quality?
- Is it due to pricing?
- Is it due to good customer service ?
- Is it due to company reputation?
- Is it something more?

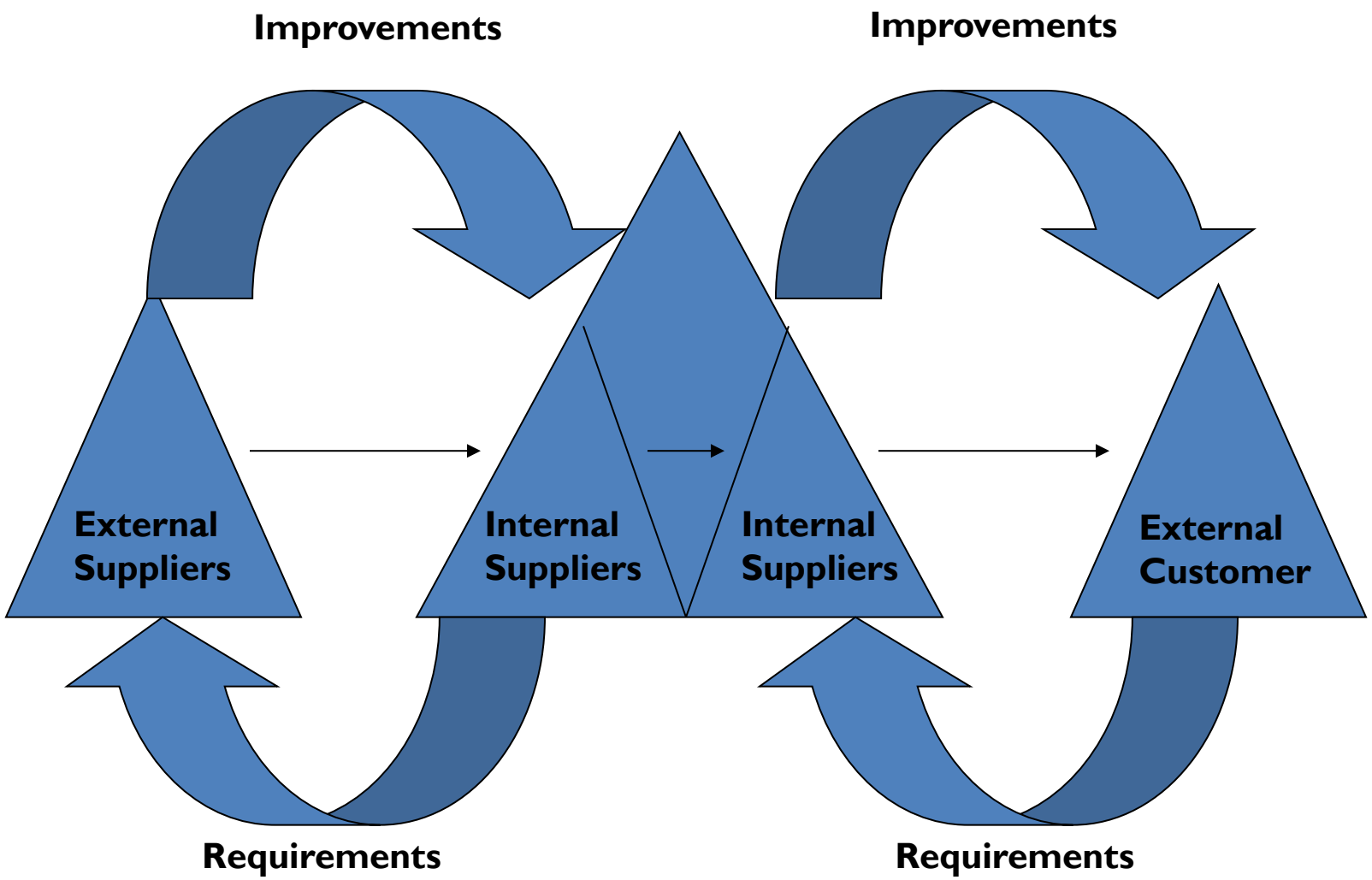




Integrated Quality



Internal / External Customer



Internal customer / Supplier relationships

- Questions asked by people to their internal customers
  - **What do you need from me?**
  - **What do you do with my output?**
  - **Are there any gaps between what you need and what you get?**

Good team-work and inter - departmental harmony is required. Also the leaders role in supervising the **internal customer-supplier chain**.





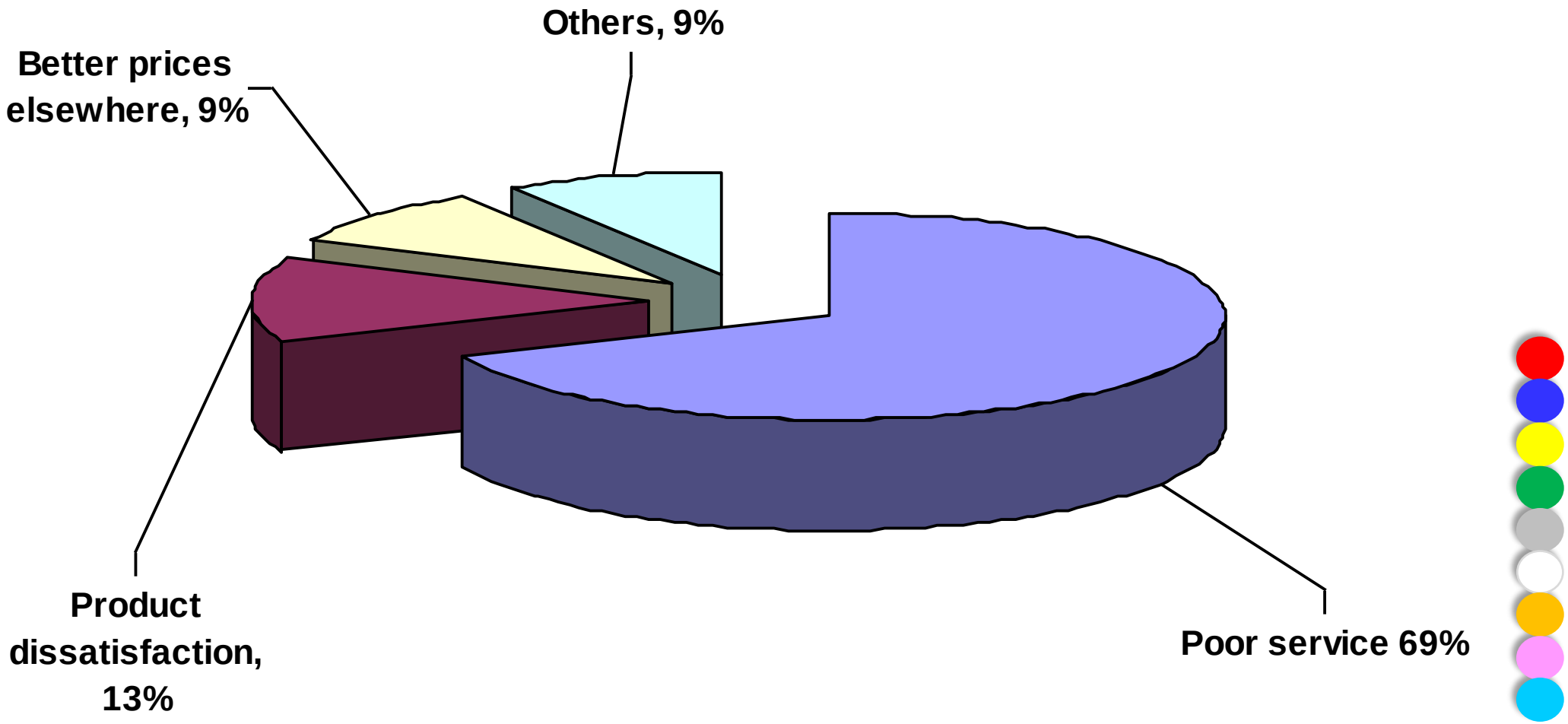
Service organizations and customer satisfaction

- Difficult but not impossible
- Determine key performance indicators
- Set Targets – SMART goals
- Make teams
- Analyze them on a weekly basis
- Standardize
- Sustain





Why Customers leave?



Difference between…..

Good service



Bad service



Excellent service



Good Customer Service



Good service is when the customer gets treatment that meets his/her expectations.



Customer Expectation



What Customer receives

Bad Customer Service



Bad Service is when customer gets treatment which is less than his/her expectations



Customer Expectation



What Customer receives

Excellent Customer Service



When the customer gets a little more than what he/she expected, Good Service becomes Excellent Service



Customer Expectation



+



What Customer receives

Categories of Customer

Angry customers



unreasonable requests and demand
Ex. Deadline, Price, 1+1

1.Stay calm.
2.Be an active listener.
3.Personalize the interaction.
4.Acknowledge your customer's emotions.
5.Use positive language.
6.Restate what they told you.
7.Build trust.
8.Thank them.

Demanding customers



They think their needs and demands are superior to everybody else's. They scream, complain, abuse and may often get physical to get what they want.

Passive customers



do not feel any loyalty towards your organization and are happy to switch brands or products as soon as a better option comes along.

Customer perception of quality

American Society for Quality Survey shows that important factors that influence purchasing are

1. PERFORMANCE – Fitness for use
2. FEATURES – Secondary characteristics
3. SERVICE – Good service at right time
4. WARRANTY – Organizations public promise of a quality product
5. PRICE – High quality products at lower prices
6. REPUTATION – willing to buy from reputed organization





Identifying customer needs

1. Customer surveys, Market research programs and studies
2. Tracking customer complaints by letters and telephone
3. Customer meetings
4. Information on competitors products
5. Personal visits to customer locations
6. Employees with special knowledge of the customer





Customer Complaints

A customer complaint may be defined as an expression of dissatisfaction with a product or service, either orally or in writing, from an internal or external customer.

Why is it necessary?

- To discover customer dissatisfaction,
- To identify customer needs,
- To discover relative priorities of quality,
- To compare performance with the competition and
- To determine opportunities, for improvement.





Various tools for collecting customer feedback or customer complaint

1. Customer cards or comment cards – question and answer type, normally attached to the warranty card.

2. Customer questionnaire – survey through website, E-mail, telephone.

3. Post–transaction surveys – conducted immediately after a customer service interaction.

4. Reports or Feedback cards – physical paper cards or forms with one or more survey questions designed to collect customer feedback.

Various tools for collecting customer feedback or customer complaint

5. Social media – social networks, online communities, blogs, and discussion boards can be used to collect customer feedback.

6. Toll-free telephone numbers – Effective tool for receiving customer feedback.

7. Customer visits – visit the customer's place to gather information and feedback on the product.

Customer satisfaction / dissatisfaction feedback

- Customer feedback has to be continuously sought and monitored - not one-time only!

  (Pro-active! Complaints are a reactive method of finding out there is a problem)

- Customer feedback can be related to Manufacturer
- Performance comparison with competitors can be known
- Customers needs can be identified
- Relative priorities of quality can be obtained from the feedback
- Areas for improvement can be noted.





Customers - Handle with care

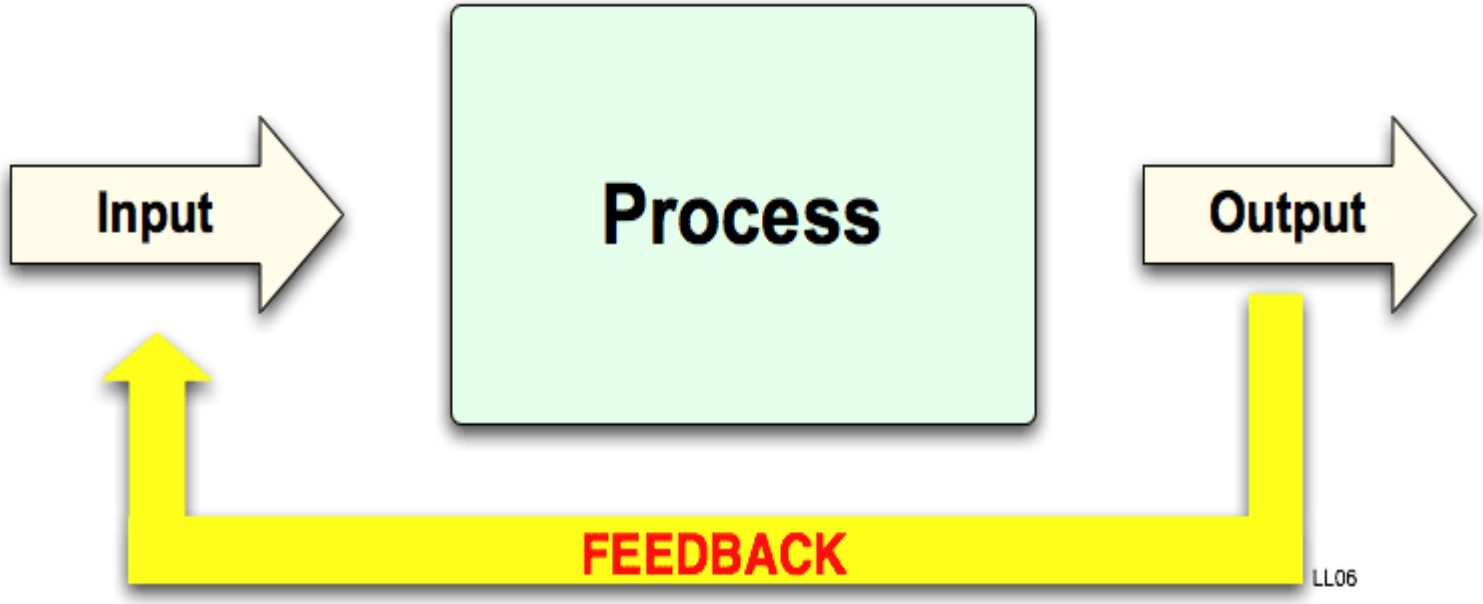
- Employers don't pay wages but it is the customer who pays the wages!
- So take good care of your customers.
- Customer-care centers not just profit-centers!
- The entire organization must in effect revolve around the customer – whether the customer is being well served and if he is really pleased, contented and satisfied with the service you have to offer.





Continuous Process Improvement

- Create a system for constant and continuous improvement



Continuous Process Improvement

Understand the Process: Variations are Major Enemies

- Detecting mode (heavy dependence on inspection/testing/control):

  Fire fighting is not effective

- Preventive mode (emphasis on "do it right the first time and all the time):

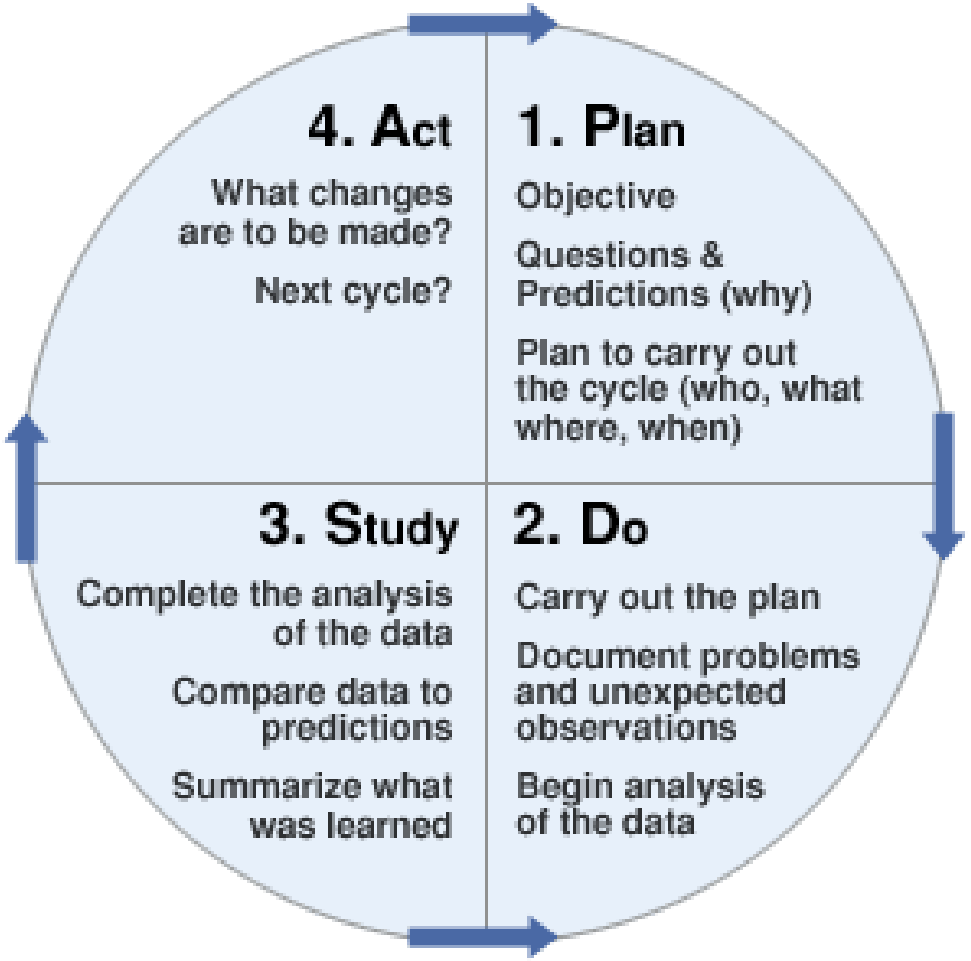
  Use FEEDBACK to take corrective actions so that the process can be continuously improved.





Continuous Process Improvement

Deming's PDSA Cycle: Embedded in the Continuing System



Continuous Process Improvement

Reasons for Collecting Data

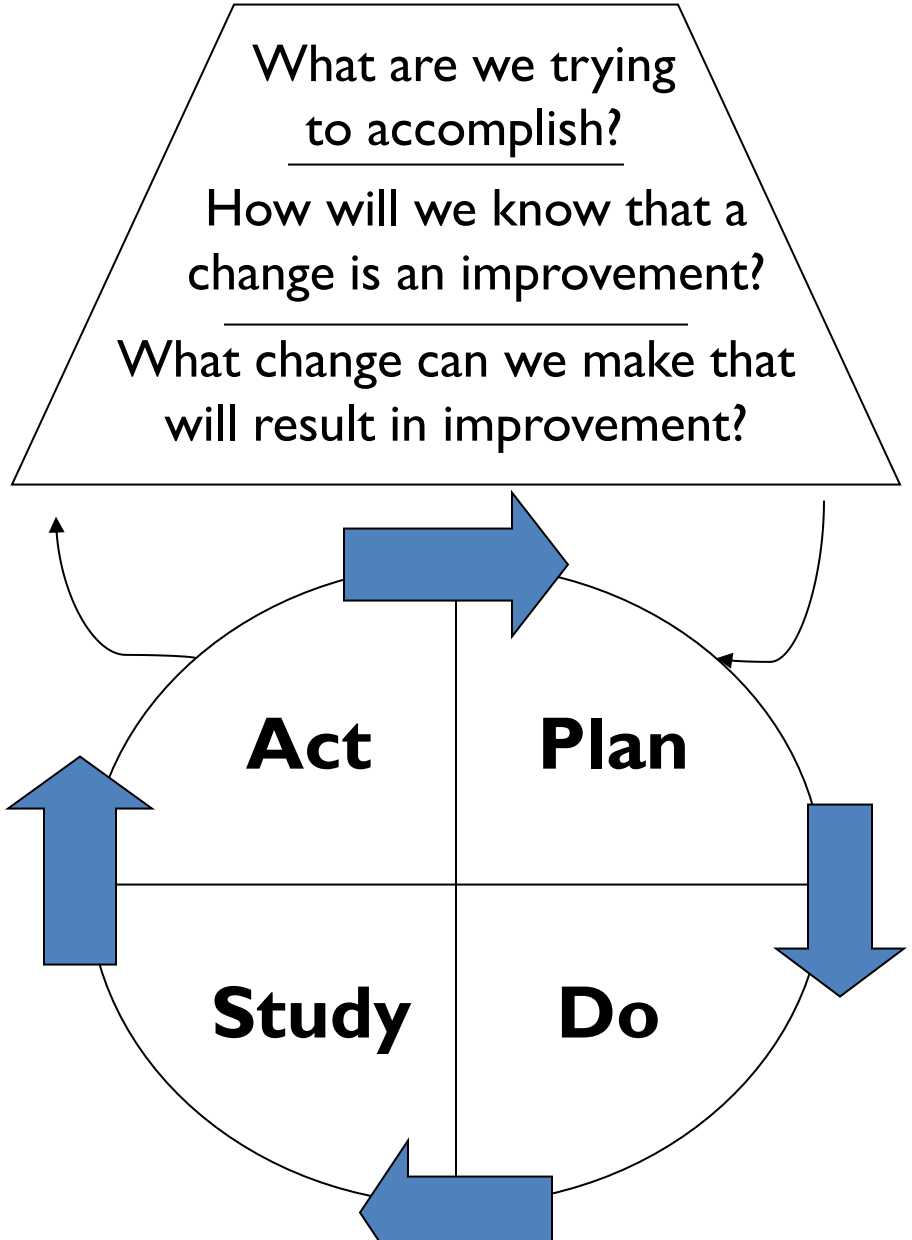
- Provide objective information
- Increase info about the change (Positive/negative)
- Give some expectation of the change impact
- Show if the gains are retained
- Involve the staff in the changes
- Provide economic impacts





Continuous Process Improvement

Model for Improvement



Continuous Process Improvement

Problem Solving: A Continuous Effort

1. Identify problems as an opportunity
2. Analyze the problem: to find root causes
3. Develop optimal and cost effective solutions
4. Implement changes: system wide
5. Study the results: worked or not? Need adjustment?
6. Standardize solution: Keep problems from reoccurring.





Continuous Process Improvement

Process Improvement Strategy: Finding Cost Effective Solution

- Control existing process: *i.e.*, statistical process control (SPC)
- Optimize the process: *i.e.*, design of experiment (DOE)
- Change the system: hardware, software and brainware
- New system: machine, human, materials, method





Continuous Process Improvement

Process Improvement Strategy: Finding Cost Effective Solution

- Build a quality (continuous improvement) culture. Positive and Constructive.
- Drive out fear.
- Provide organization-structural assurance for continuing feedback system.
- Respond to customers needs and solve problems timely.
- Standardize the procedure and continuously improve it.

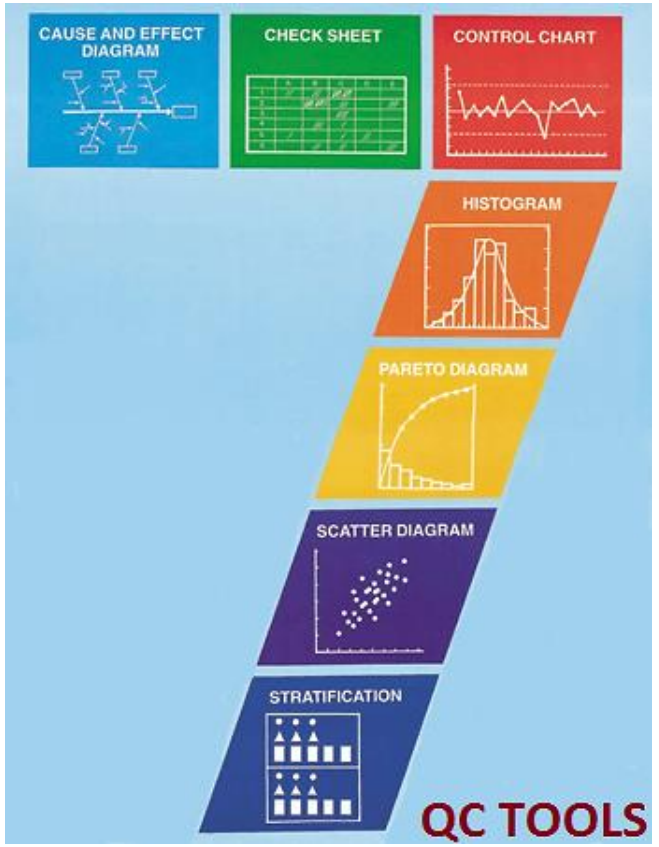




Quality Tools

- QC tools are the set/combination of 7 graphical and statistical techniques for identifying & solving the issue-related quality





1. Check-sheet

- The check sheet is the form/document used to collect data in real-time at the location where the data is generated. The data it captures can be **quantitative** or **qualitative**.

Uses:

- To quantify the types of problems that are occurring
- To quantify the defects by type
- To quantify defects by location

SAMPLE-1

DEFECT CHECKSHEET					Department: Administration Task: Typing
Mistake	March 2020				Total
	1st Week	2nd Week	3rd Week	4th Week	
Centering	||	|||	|||	|	9
Spelling	|| 𝍸	| 𝍸 𝍸	𝍸	|| 𝍸	30
Punctuation	||| 𝍸 𝍸	|| 𝍸 𝍸	|| 𝍸	| 𝍸 𝍸	43
Missed Paragraph		|	||	|	4
Wrong number	|||	||	||||	|	10
Total	25	29	21	21	96

SAMPLE-2

CAUSE CHECKSHEET			Item Group: Suppliers Defective bills Period: 1st April to 30th June, 2019		
Cause of return of bills by Account	Run				Total
	Run # 1	Run # 2	Run # 3	Run # 4	
Purchase order not released	5	5	6	7	23
Form 31 not signed by the supplier	1		2	1	4
Rate difference	3	1	2	1	7
Packing and forwarding charges not per P.O	–	–	1	–	1
Freight bill not attached	1	–	–	1	2
Totaling errors	2	–	–	–	2
Total (No. of defective bills)	12	6	10*	9*	39

SAMPLE-3

INSPECTION CHECKSHEET			Department: Maintenance Task: Preventive inspection Month: June 2019		
Check Point	Week				Remarks
	1st	2nd	3rd	4th	
Check oil level	√	√	√	√	
Clean battery top	√	√	√	√	
Check electrolyte level	OK	OK	OK	Filled	
Check oil leakage	√	√	√	√	
Check break	√	√	√	√	
Check clutch	OK	OK	OK	Adjusted	
Tick off of write OK is satisfied					
				Signature (Maint.	

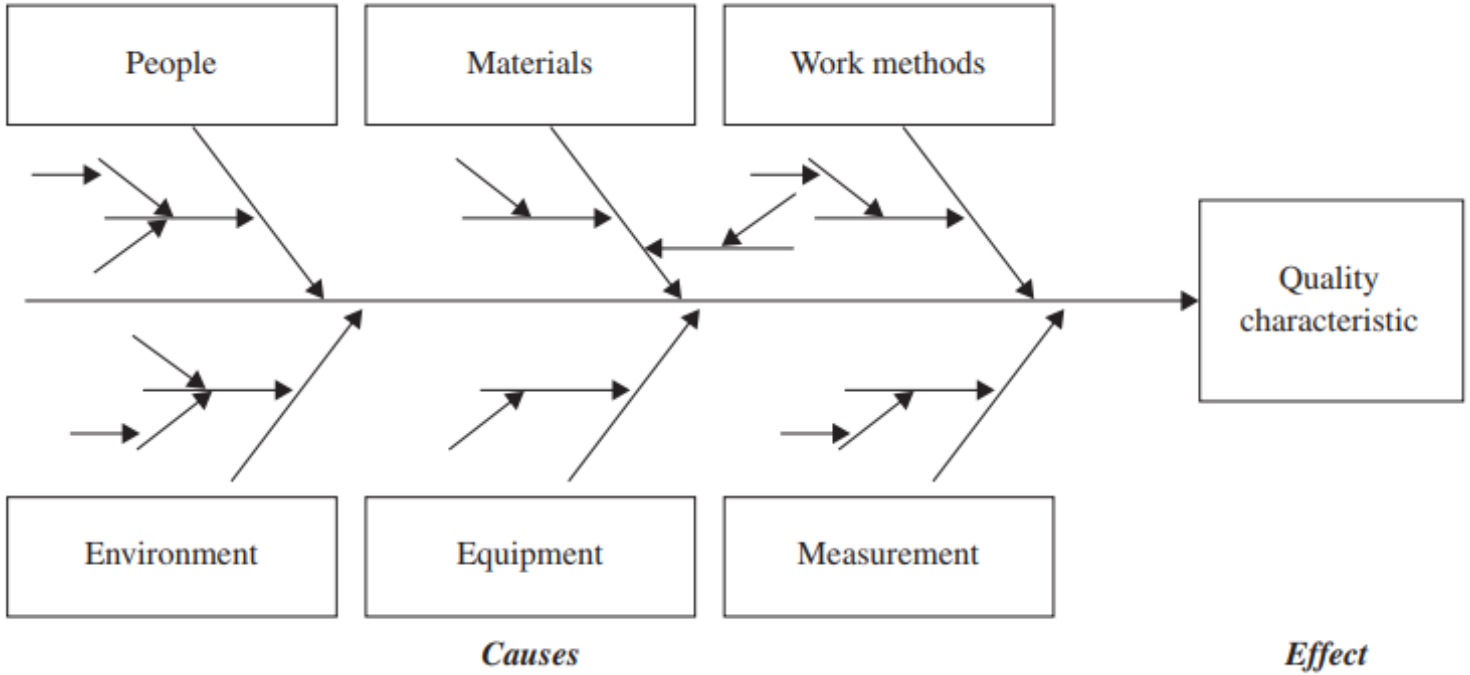
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2. Fishbone diagram

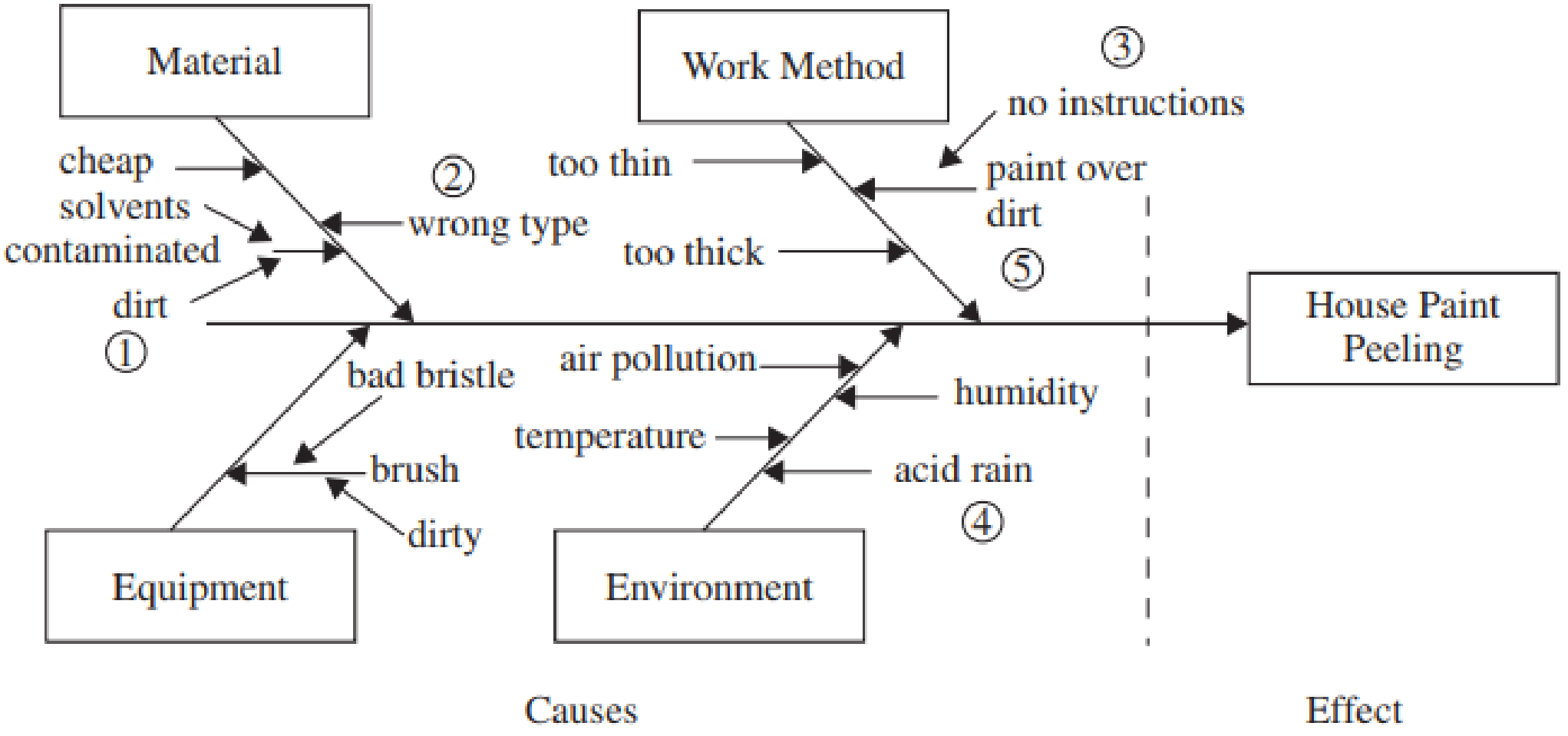
- It also known as the Cause & Effect Diagram or Ishikawa Diagram (Mr. Kaoru Ishikawa)
- It identifies many possible causes for an effect or problem and sorts ideas into useful categories (6M or 5M+1E)

Uses:

- Identify the possible root causes for any effect/problem
- It is used to structure a brainstorming session



2. Fishbone Cond...



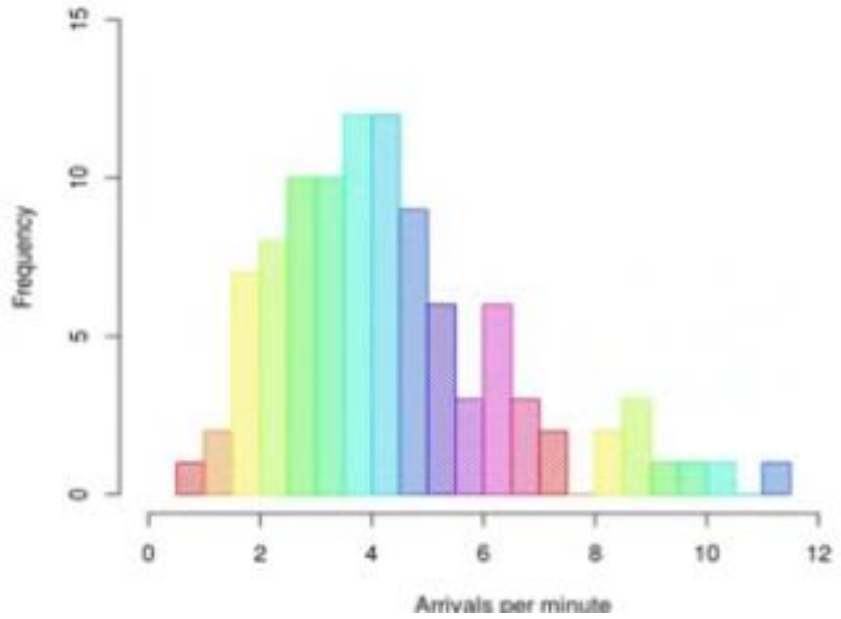
Cause-and-Effect Diagram of House Paint Peeling

3. Histogram

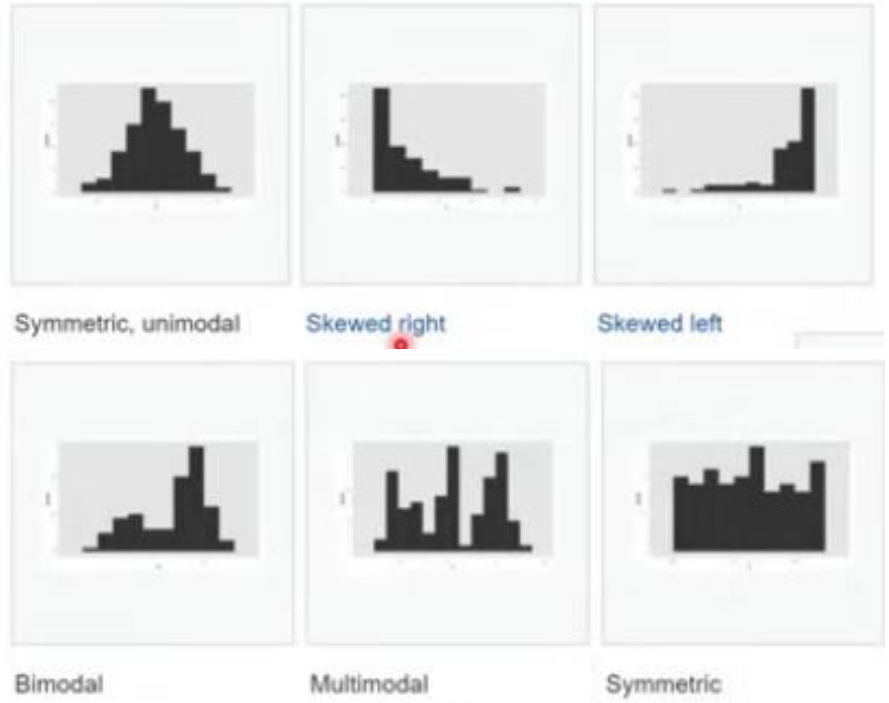
- Karl Pearson introduced it and it is a bar graph representing the frequency distribution on each bars or How often each different value in a set of data occurs

Uses:

- Measure the current pattern and extent of variation.
- Assess correct conformance to specifications and the scope for improvements
- Determine whether or not the process is in the state of control



i. Histogram for late arrival



ii. Different Histogram

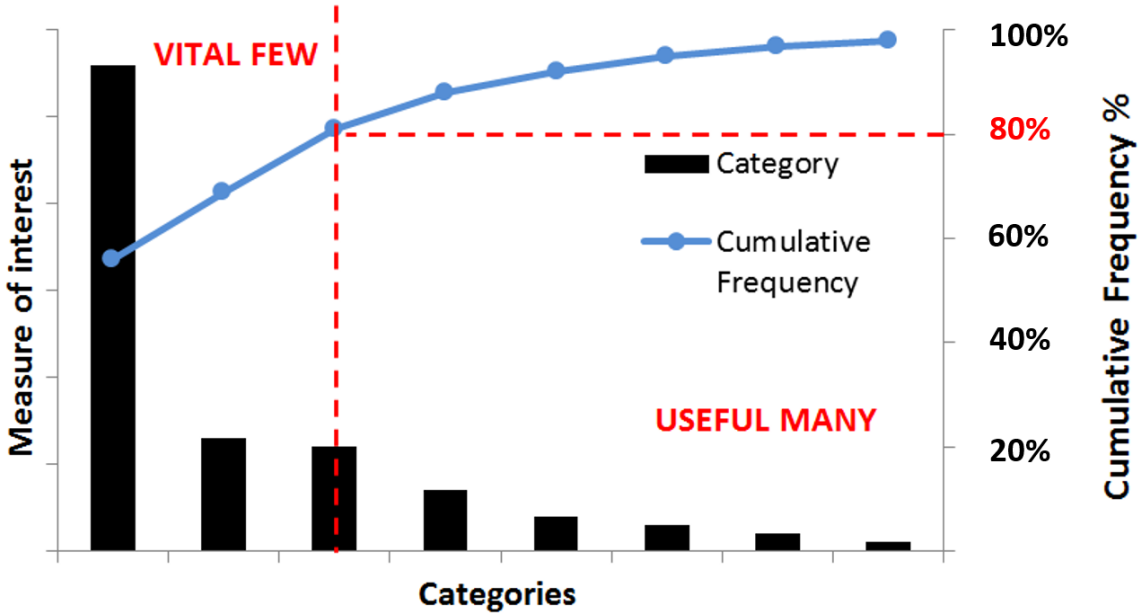
4. PARETO-Chart

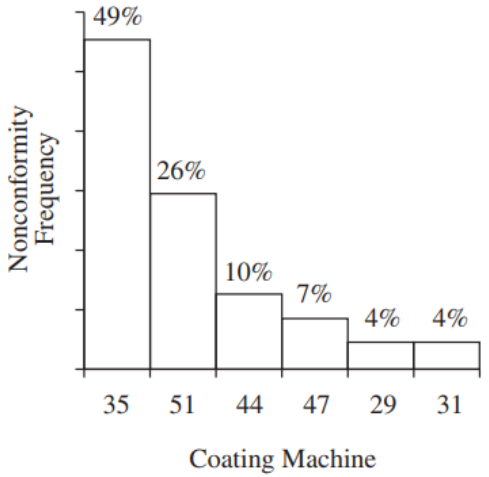
- The Pareto chart operates according to the **80-20 rule**.
- The Pareto Principle is named after the economist **Vilfredo Pareto**, who specifies that **80% of consequences (results) come from 20% of the causes (actions).**
- This rule assumes that in any process, 80% of a process's or system's problems are caused by 20% of major factors, often referred to as the "vital few." The remaining 20% of problems are caused by 80% of minor factors.
- A combination of a bar and line graph, the Pareto chart depicts individual values in descending order using bars, while the cumulative total is represented by the line.

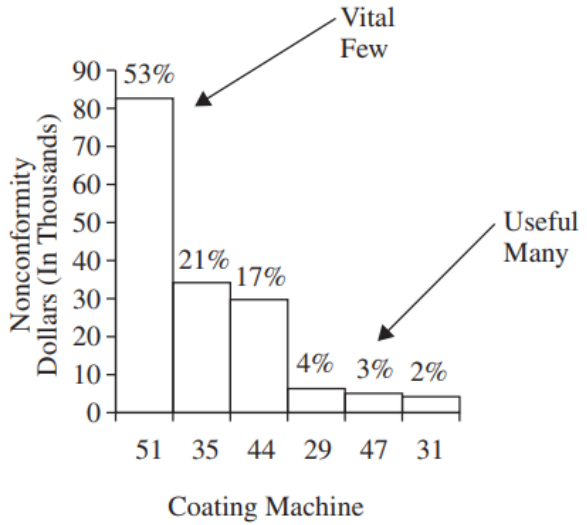
Uses:

- Distinguish between vital few and vital many
- Display relative importance of the causes of a problem
- Helps to focus on causes that will have the greatest impact when solved

4. PARETO-Chart Cond...

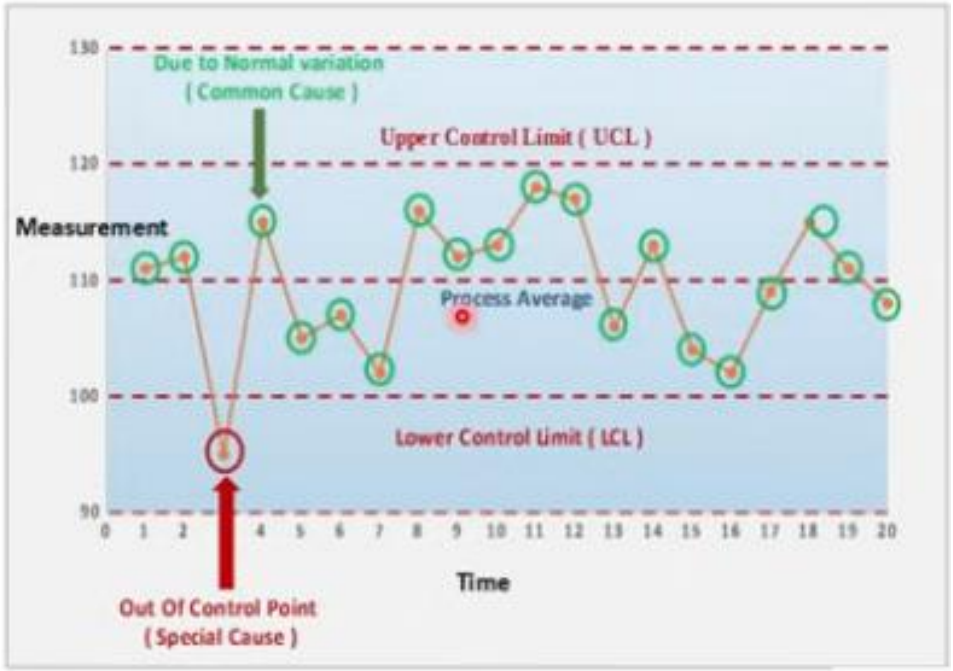






5. Control Chart

- It is also known as Shewhart charts (A Shewhart) or process-behaviour chart
- It is a statistical process control tool used to determine if a manufacturing or business process is in a state of control
- The purpose of a control chart is to determine if the process is stable and capable within current conditions
- It helps in predicting process performance, understanding the various production patterns and studying how a process changes or shifts from normally specified control limits over some time
- In the control chart, data are plotted against time in X-axis control chart will always have a central line, an upper line for the upper control limit and a lower line for the lower control limit. These lines are determined from historical data
  - UCL (Upper control limit)
  - CL (Control limit)
  - LCL (Lower control limit)

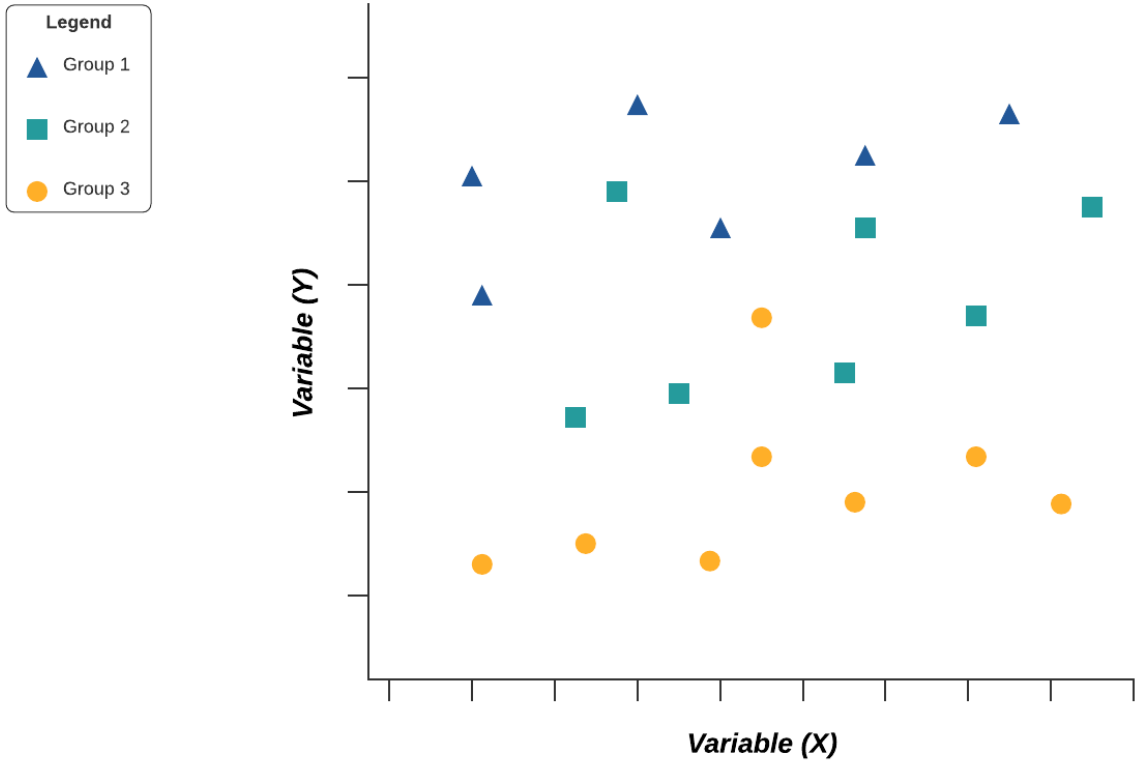


6. Scattered diagram

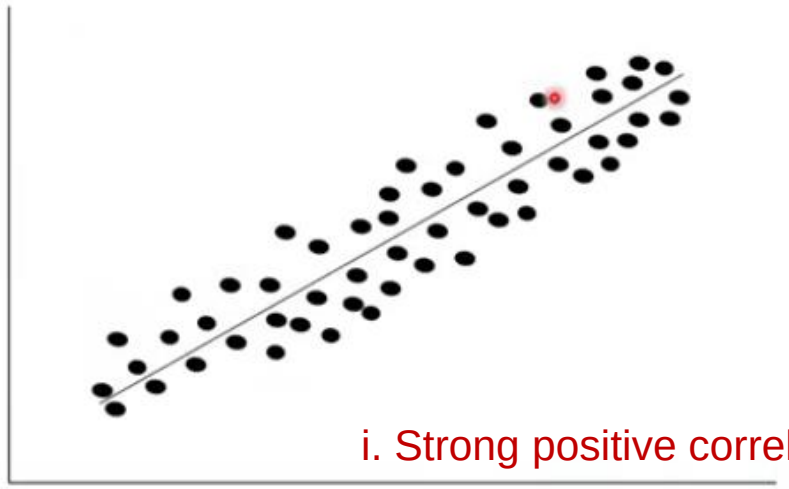
- It is also known as a **correlation diagram** which represents between two variables (between cause and effect) or between two associated sets of data which occur in pairs using Cartesian coordinates

Uses:

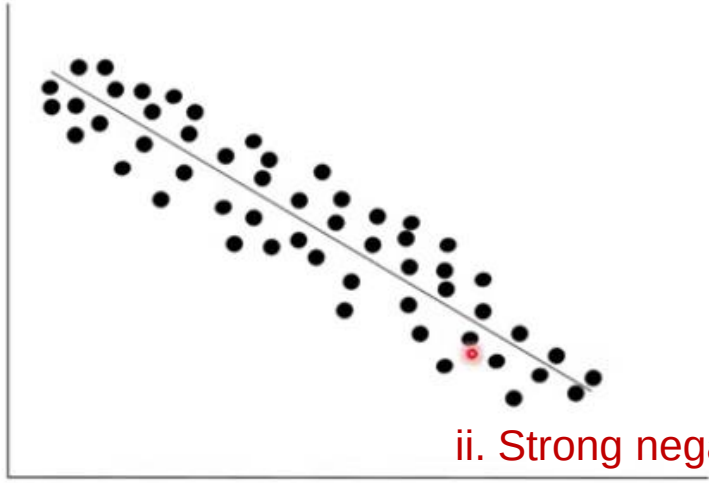
- To discover and display the relationship between two associated sets of data



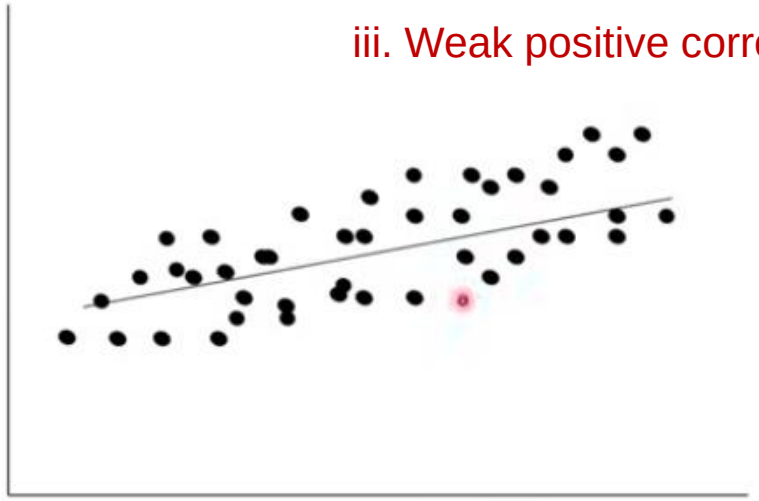
6. Scattered diagram Cond…



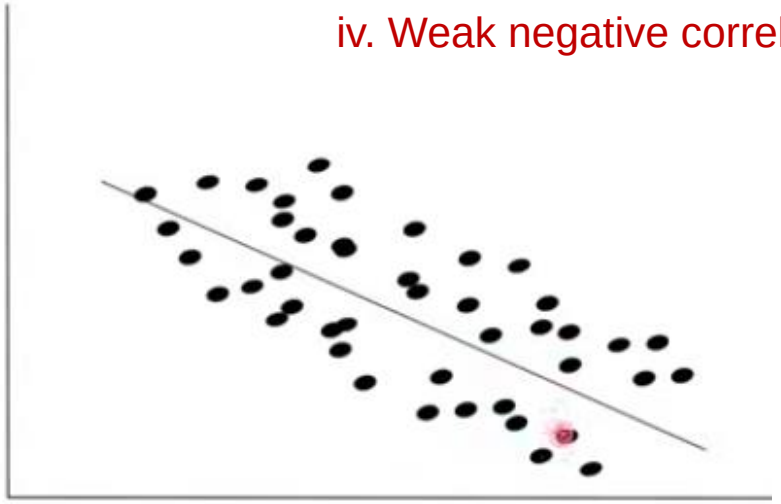
i. Strong positive correlation



ii. Strong negative correlation



iii. Weak positive correlation



iv. Weak negative correlation

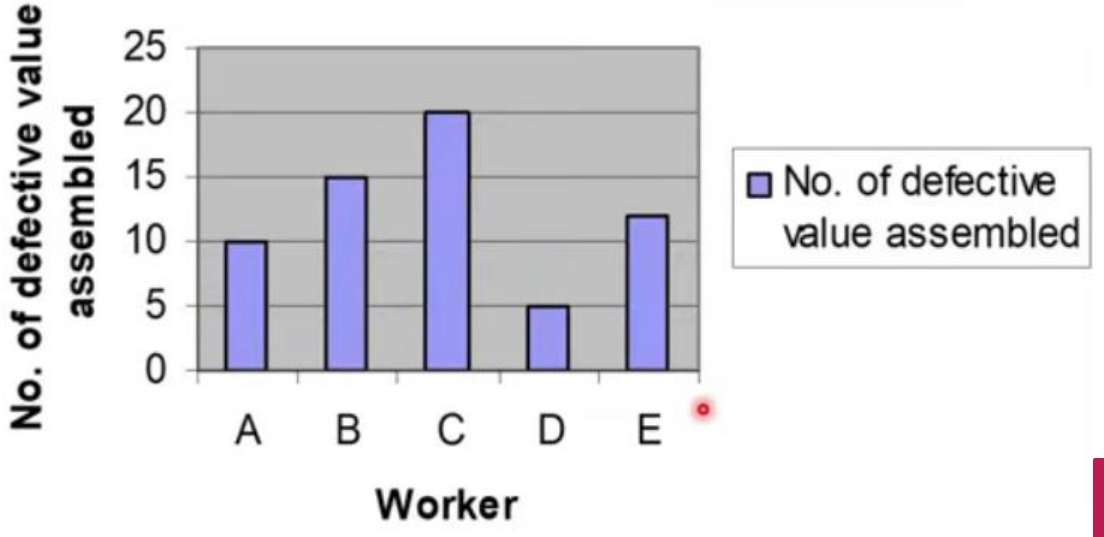
7. Stratification

- It is a method of dividing data into sub-categories and classifying data based on group, division, class or levels that helps in deriving meaningful information to understand an existing problem.

- For example **Customer complaints** may be segregated by:
  - **Nature of complaints** (e.g. defective product, delayed delivery, short/excess quality, damaged product, incorrect paperwork, incorrect shipment etc.)
  - **Departments responsible** (e.g. design, manufacturing, distribution, quality assurance sales and service etc.)

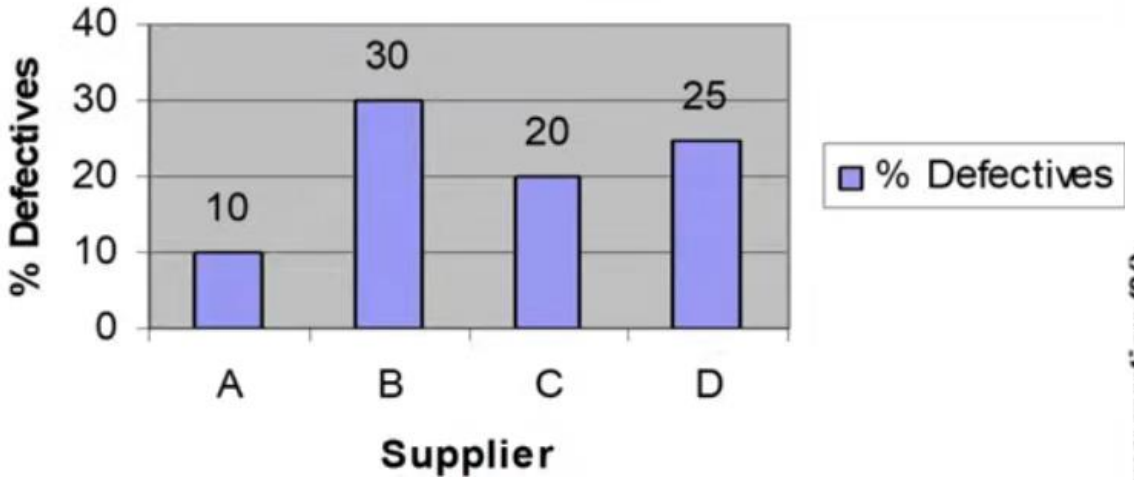
Uses:

- Separate data into groups and thereby draw meaningful and correct inferences from the data
- Diagnose and localize problems

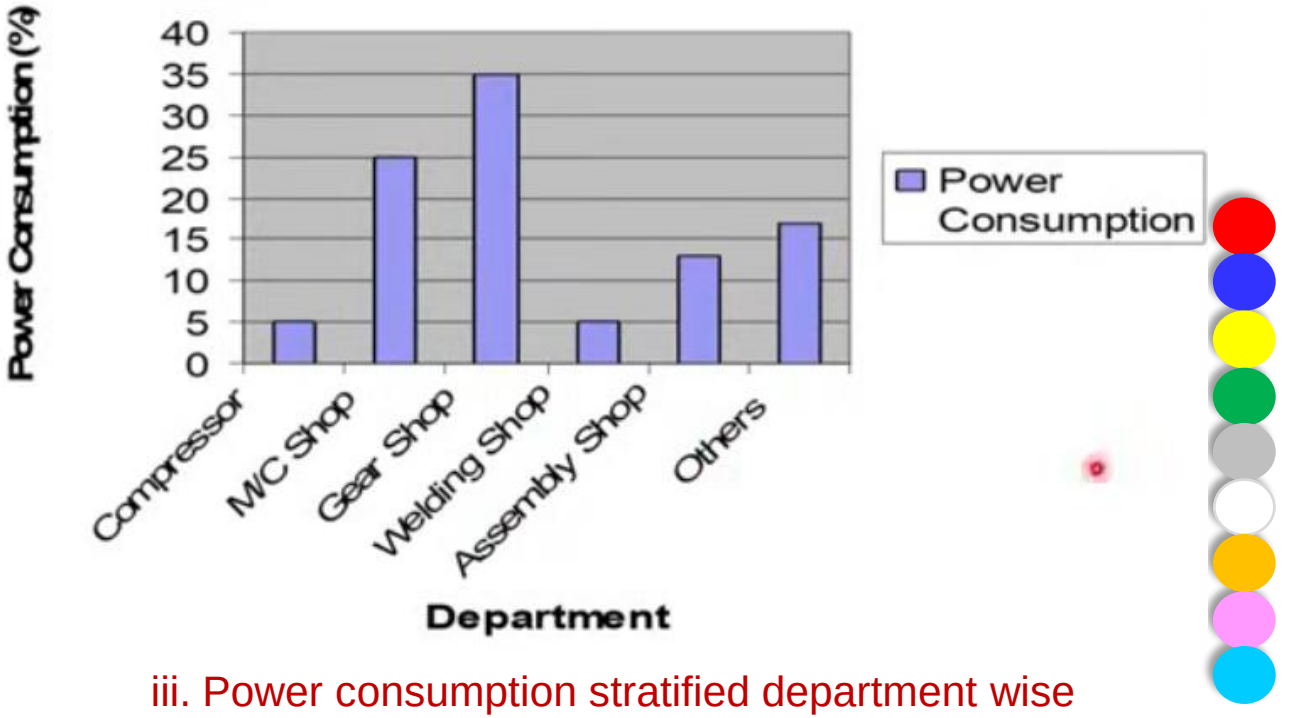


i. Defective values stratified assembly worker-wise

7. Stratification Cond…

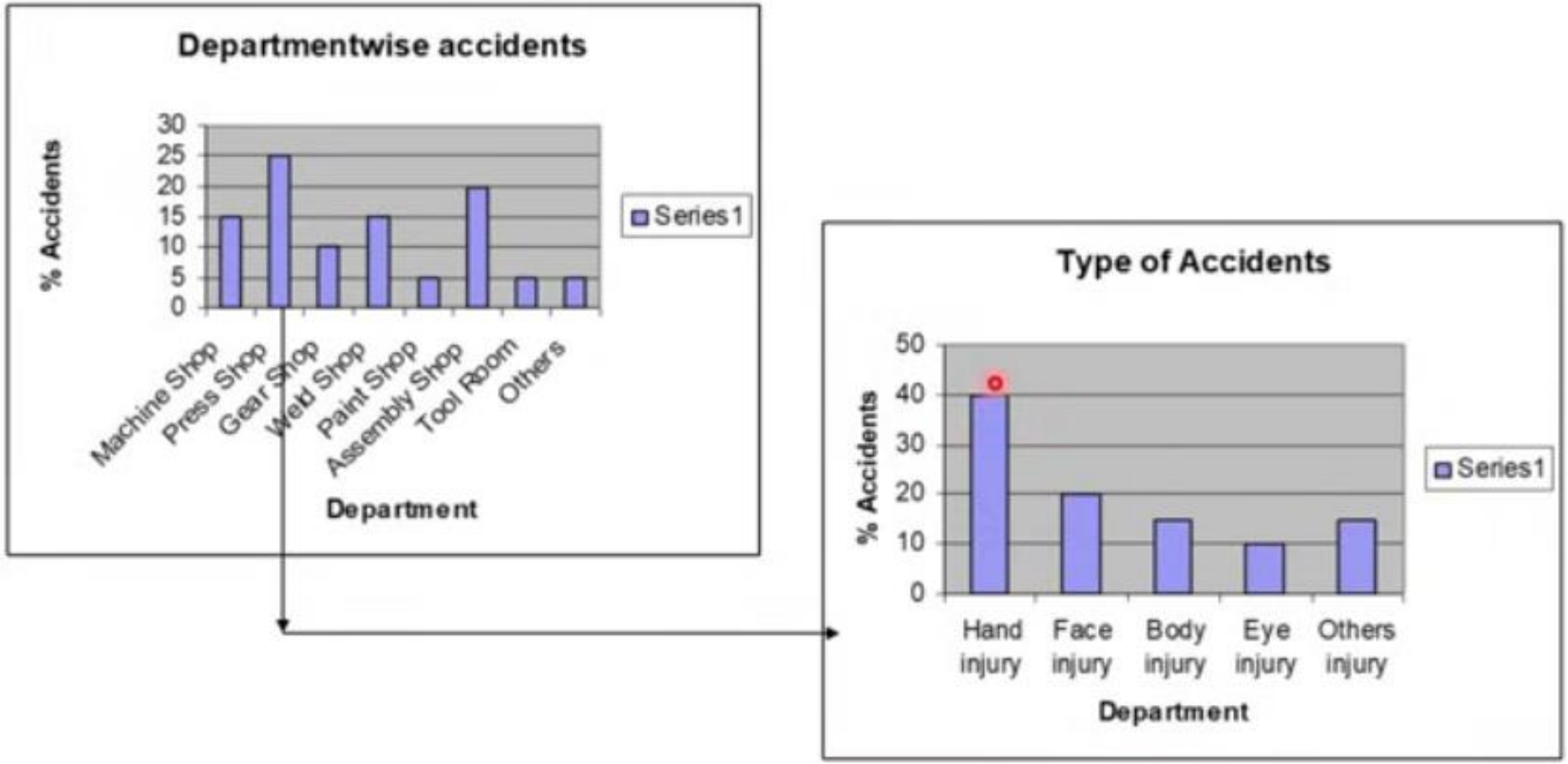


ii. % defectives stratified supplier worker-wise



iii. Power consumption stratified department wise

7. Stratification Cond…

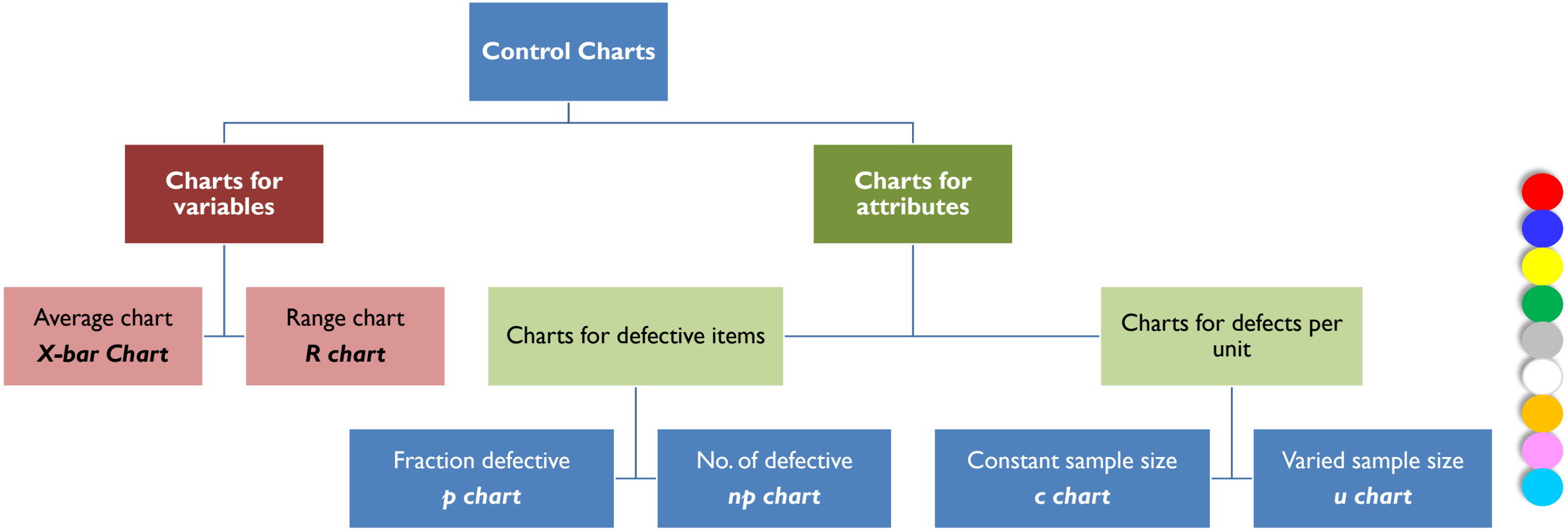


iv. Accidents stratified department-wise and types of accidents

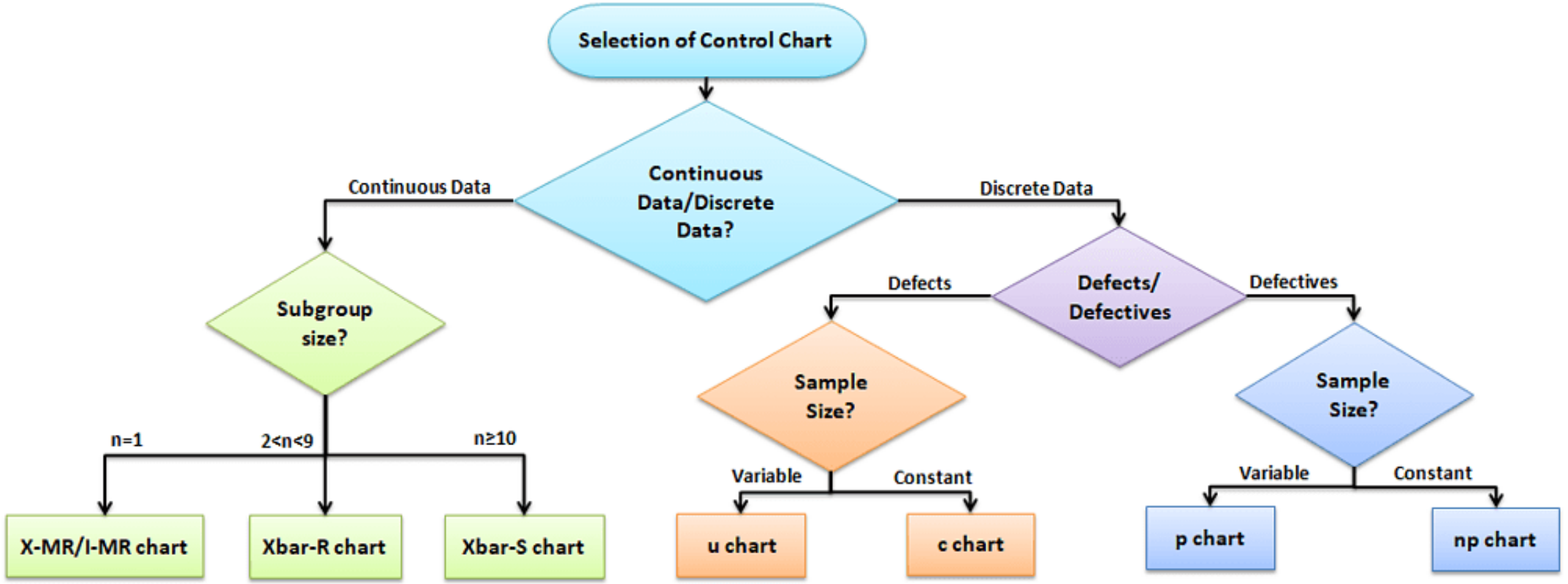
Types of Control Charts (SPC)

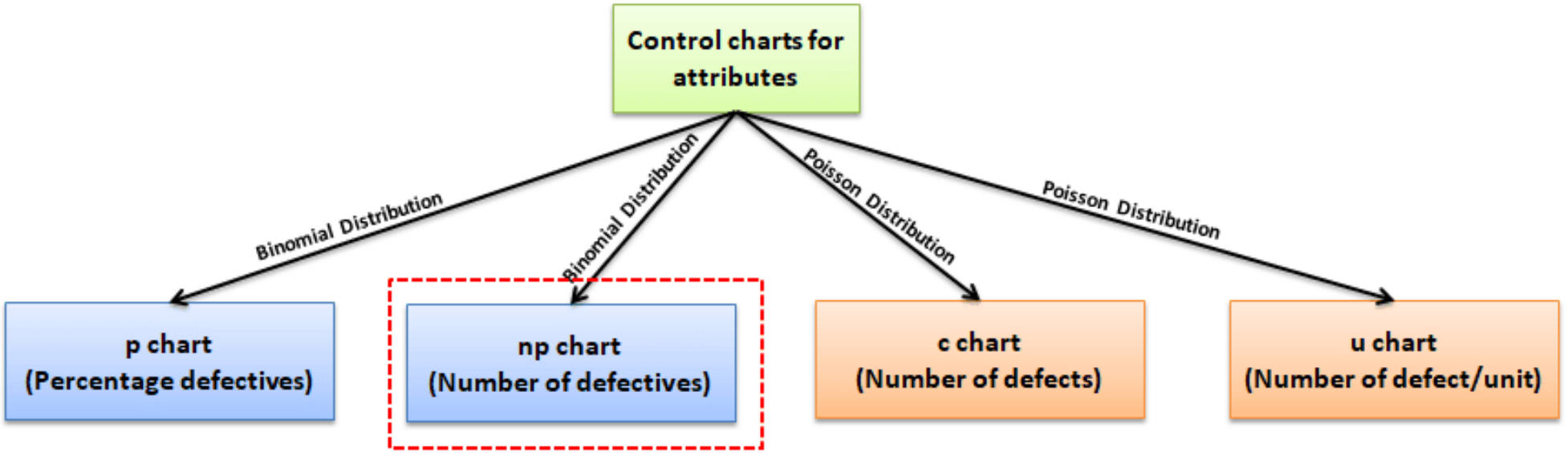
Two broad categories of charts exist, based on whether the data being monitored is "**variable**" or "**attribute**" in nature.

- Variable control charts plot measurable process data in a time-ordered sequence.
- Attribute control charts plot data that is counted instead of being measured.



How to select control charts?




Control charts for attributes
Binomial Distribution
Binomial Distribution
Poisson Distribution
Poisson Distribution
p chart (Percentage defectives)
np chart (Number of defectives)
c chart (Number of defects)
u chart (Number of defect/unit)

X bar chart

- X-Charts present variable data. Each data point is the statistical mean of a subgroup of values.
- The statistical principles used in calculating the UCL and LCL are based on normal distribution.
- This type of chart graphs the means (or averages) of a set of samples, plotted in order to monitor the mean of a variable.
- For example: the length of steel rods, the weight of bags of compound, the intensity of laser beams, etc..
- In constructing this chart, samples of process outputs are taken at regular intervals, the means of each set of samples are calculated and graphed onto the X bar control chart.
- This chart can be utilized to determine the actual process mean, versus a nominal process mean and will demonstrate if the mean output of the process is changing over time.





R chart

- This type of chart demonstrates the variability within a process.
- It is suited to processes where the sample sizes are relatively small, for example <10.
- Sets of sample data are recorded from a process for the particular quality characteristic being monitored.
- For each set of date the difference between the smallest and largest readings are recorded. This is the range “R” of the set of data.
- The ranges are now recorded onto a control chart.
- The center line is the averages of all the ranges.

Problems on Xbar and Rbar

Example: You are given the values of sample means and range for 10 samples of size 5 each shown in the table:

Take A_2 = 0.58, D_3=0 and D_4=2.11 respectively. Draw the charts for means and Ranges, also comment on the stages of control of the process

Sample no (n)	1	2	3	4	5	6	7	8	9	10
Mean ($\bar{X}$)	43	49	35	44	45	37	51	46	43	47
Range (R)	5	6	5	7	6	4	8	6	4	6

Average means, $\bar{\bar{X}} = \frac{\sum \bar{X}}{n} = \frac{(43+49+35+44+45+37+51+46+43+47)}{10} = \mathbf{44.6}$

Average Range $\bar{R} = \frac{\sum R}{n} = \frac{(5+6+5+7+6+4+8+6+4+6)}{10} = \mathbf{5.7}$

Problems on Xbar and Rbar

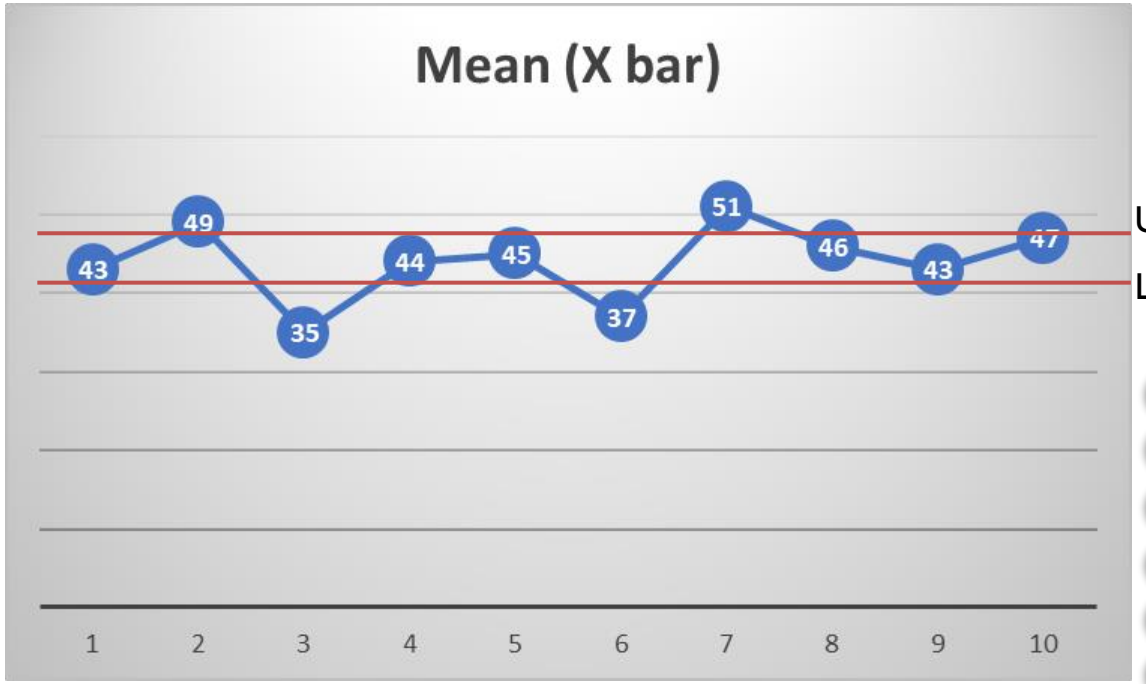
Control limits for $\bar{\bar{X}}$ chart

Upper control limit

$$\text{UCL} = \bar{\bar{X}} + A_2\bar{R} = 44.6 + (0.58 \times 5.7) = \mathbf{47.689}$$

Lower control limit

$$\text{LCL} = \bar{\bar{X}} - A_2\bar{R} = 44.6 - (0.58 \times 5.7) = \mathbf{41.311}$$



Comments: According to $\bar{X}$ chart, there are 2 points above UCL, i.e., 49, & 51 and there are 2 points below LCL, i.e., 35 and 37 indicating the process is out of control.

Problems on Xbar and Rbar

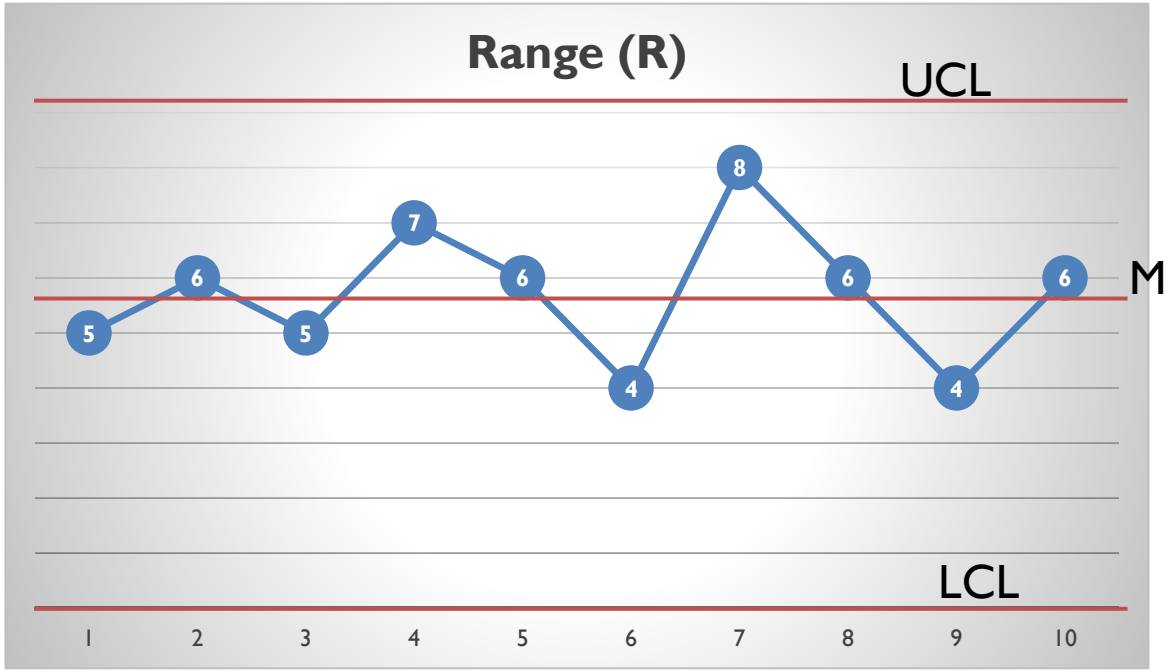
Control limits for $\bar{R}$ chart

Upper control limit

$$UCL=D_4\bar{R} = 2.11\times5.7 = 12.03$$

Lower control limit

$$LCL=D_3\bar{R} = 0\times5.7 = 0$$



Comments: According to $\bar{R}$ chart, all the sample points lies within limits, it means process in in control.

Problems on Xbar and Rbar

Sample	Obsevation				
	1	2	3	4	5
1	0.1253	0.1262	0.1254	0.124	0.123
2	0.1242	0.1247	0.1251	0.1238	0.1241
3	0.1225	0.1258	0.1229	0.1242	0.1255
4	0.1249	0.1259	0.1249	0.124	0.1257
5	0.1245	0.1252	0.1261	0.1238	0.1225
6	0.1273	0.1234	0.1248	0.1241	0.126
7	0.1226	0.1239	0.1227	0.1252	0.1259
8	0.1244	0.1238	0.1254	0.1261	0.126
9	0.1236	0.1262	0.125	0.1247	0.125
10	0.1251	0.1264	0.1233	0.1233	0.1246
11	0.1253	0.1248	0.137	0.1252	0.1226
12	0.1232	0.1251	0.1259	0.1263	0.1257
13	0.1231	0.1242	0.1256	0.1252	0.1257
14	0.1256	0.124	0.1246	0.125	0.1252
15	0.1243	0.124	0.1239	0.1262	0.1246

SAMPLE SIZE *n*	*A2*	*D3*	*D4*
2	1.88	0	3.27
3	1.02	0	2.57
4	0.73	0	2.28
5	0.58	0	2.11
6	0.48	0	2.00
7	0.42	0.08	1.92
8	0.37	0.14	1.86
9	0.34	0.18	1.82
10	0.31	0.22	1.78
11	0.29	0.26	1.74
12	0.27	0.28	1.72





p chart

- P charts are utilized where there is a pass / fail determination on a unit inspected.
- P-Charts are used for data that is counted.
- The p chart will show if the proportion defective within a process changes over the sampling period (the p indicates the portion of successes).
- In the p chart the sample size can vary over time.
- Each data point is a ratio of its own discrete sample set. P-Charts can display fraction nonconforming or percent nonconforming data values.
- The statistical principles used in calculating the UCL and LCL are based on binomial distribution.

np chart

- np-Charts are used to present the number of non-conforming or conforming items.
- The sample subgroup size must be uniform.
- In the np chart, the sample size needs to stay constant over the sampling period.
- In the np chart, the number non-conforming is recorded onto the control rather than the fraction non-conforming.
- Some process operators are more comfortable plotting the number than the fraction of non-conformances.
- The statistical principles of calculating the UCL and LCL are based on binomial distribution.





u chart

- The “u” and “c” control charts are applied when monitoring and controlling count data in the form of 1,2,3, …. i.e. specific numbers.
- An example of such data is the number of defects in a batch of raw material or the number of defects identified within a finished product.
- U charts display the average count of nonconforming items per unit when the unit size is variable.
- The statistical principles of calculating the **UCL** and **LCL** are based on the Poisson distribution.





c chart

- The c chart is used where there can be a number of defects per sample unit, and the number of samples per sampling period remains constant.
- In the u chart, again similar to the c chart, the number of defects per sample unit can be recorded, however, with the u chart, the number of samples per sampling period may vary.
- C-Charts display the number of nonconforming or conforming items within a consistent unit size or the probability of a specific number of occurrences in a specific period of time.
- The statistical principles of calculating the UCL and LCL are based on the Poisson distribution.





Tests for checking out of control

The four tests we shall used for determining when a process is out of control are as follows:

(i) One or more points outside the control limits.

(ii) 9 points in a row above or below the center line.

(iii) 6 points in a row steadily increasing or decreasing.

(iv) 14 points in a row alternating up and down (not necessarily crossing the center line).





X bar chart and R chart

$$\overline{X_i} = \frac{\sum_{i=1}^{n} X_i}{n}$$

$\overline{X_i}$ = mean of the ith sample; n = sample size; X_i = i^{th} data

$$R_i = X_{\max(i)} - X_{\min(i)}$$

R_i = range of ith sample
$X_{max(i)}$ = maximum value of the data in i^{th} sample
$X_{min(i)}$ = minimum value of the data in i^{th} sample

$$\overline{R} = \frac{\sum_{i=1}^{g} R_i}{g}$$

$\overline{R}$ = mean of g samples

$$\overline{\overline{X}} = \frac{\sum_{i=1}^{g} \overline{X_i}}{g} = CL_{\overline{X}}$$ **(Centre line for X bar chart)**

$\overline{\overline{X}}$ = mean of mean of g samples
g = number of samples

X bar chart and R chart

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \bar{R}$$ **(Upper control limit for X bar chart)**

$$A_2 = \frac{3}{d_2 \sqrt{n}} =$$ Parameter depends on sample size

value of A_2 can be directly obtained from the standard tables

$$LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \bar{R}$$ **(Lower control limit for X bar chart)**

$$UCL_R = D_4 \bar{R} \text{ where } D_4 = \bar{R}\left(1 + \frac{3d_3}{d_2}\right)$$ **(Upper control limit for R chart)**

$$UCL_R = D_3 \bar{R} \text{ where } D_3 = \bar{R}\left(1 - \frac{3d_3}{d_2}\right)$$ **(Lower control limit for R chart)**





X bar chart and R chart

$\bar{X}$ & R Chart				
Subgroup Size (n)	A_2	D_3	D_4	d_2
2	1.880	0	3.267	1.128
3	1.023	0	2.574	1.693
4	0.729	0	2.282	2.059
5	0.577	0	2.114	2.326
6	0.483	0	2.004	2.534
7	0.419	0.076	1.924	2.704
8	0.373	0.136	1.864	2.847
9	0.337	0.184	1.816	2.970
10	0.308	0.223	1.777	3.078





X bar chart and R chart - example

Mean values and ranges of sample data from 20 samples (sample size = 4) are given in the table below:

S. No.	Mean of Sample	Range	S. No.	Mean of Sample	Range	S. No.	Mean of Sample	Range	S. No.	Mean of Sample	Range	S. No.	Mean of Sample	Range
1	10	4	5	9	5	9	10	4	13	12	4	17	12	4
2	15	4	6	11	6	10	11	6	14	12	3	18	15	3
3	12	5	7	11	4	11	12	5	15	11	3	19	11	3
4	11	4	8	9	4	12	13	4	16	15	4	20	10	4

X bar chart and R chart - example

Sum of mean of 20 samples = $\sum_{i=1}^{20} \overline{X} = 232$

Average of mean values of 20 samples = $\frac{\sum_{i=1}^{20} \overline{X}}{20} = 11.6$ (Center Line of X bar Chart)

Average of Ranges of 20 samples = $\frac{\sum_{i=1}^{20} R}{20} = 4.15$ (Center Line of R Chart)

Upper Control Limit of X bar chart = $11.6 + A_2\ 4.15$ ($A_2 = 0.729$ for sample size 4)

$= 14.63$

Lower Control Limit of X bar chart = $11.6 - A_2\ 4.15$ ($A_2 = 0.729$ for sample size 4)

$= 8.57$

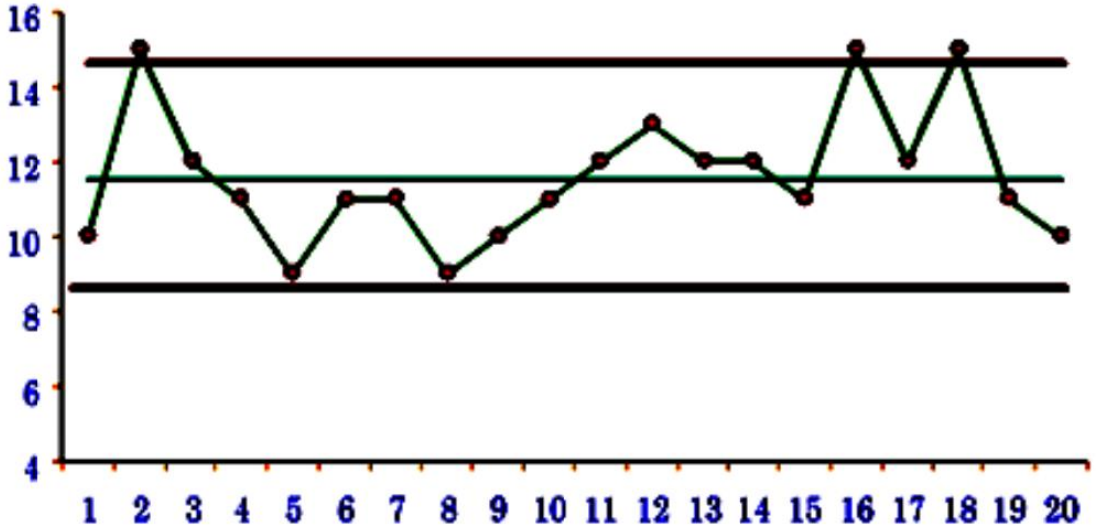
Upper Control Limit of R chart = $D_3\ 4.15$ ($D_3 = 2.282$ for sample size 4)

$= 9.47 \approx 9.5$

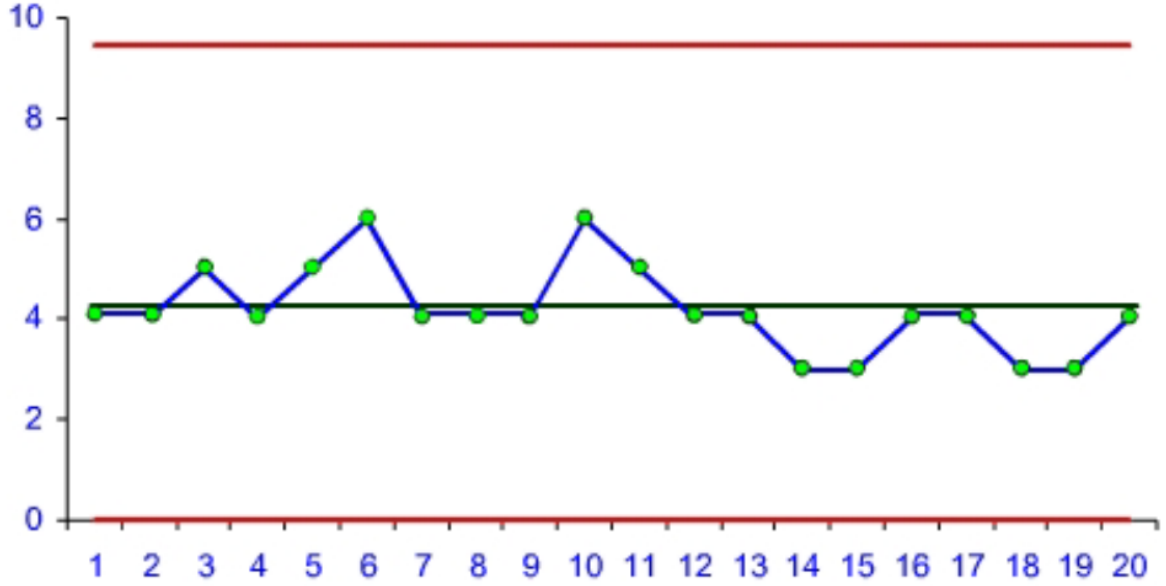
Lower Control Limit of R chart = $D_4\ 4.15$ ($D_4 = 0$ for sample size 4)

X bar chart and R chart - example

X bar chart



R chart



The following data gives the mean and range of 25 samples each consisting of four compression test results of steel forgings

Sample	$\overline{X}$	R	Sample	$\overline{X}$	R
1	55.4	3.7	14	59.2	4.1
2	58.1	4.1	15	61.1	2.5
3	56.2	6.0	16	52.8	3.2
4	55.7	2.6	17	61.11	2.4
5	51.9	3.2	18	62.4	5.3
6	59.4	6.7	19	57.9	3.2
7	62.6	7.5	20	58.6	3.7
8	64.5	4.6	21	63.3	4.0
9	55.1	3.5	22	59.7	2.1
10	57.6	2.0	23	58.2	3.1
11	52.8	4.9	24	61.6	2.6
12	51.4	6.6	25	62.3	3.4
13	53.7	3.7			

[*Ans.* (*a*) $CL_{\bar{X}} = 58.1$; $UCL_{\bar{X}} = 60.98$; $LCL_{\bar{X}} = 55.22$; $CL_R = 3.948$; $UCL_R = 9$; $LCL_R = 0$; (*c*) Process is out of control; (*d*) $CL_{\bar{X}} = 57.82$; $UCL_{\bar{X}} = 60.702$; $LCL_{\bar{X}} = 54.937$; $CL_R = 3.948$; $UCL_R = 9$; $LCL_R = 0$]

p chart

$$\bar{p} = \frac{\text{Sum of defectivess peice in all samples}}{\text{Total number of items in all samples}} = \textbf{centre line of p chart}$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \quad \text{and} \quad LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

Where n is the sample size. Sample size in p chart must be ≥ 50
Sometimes LCL in p chart becomes negative, in such cases LCL should be taken as 0





p chart

p chart for proportions of units in a category	$CL_p = \bar{p}$	**Samples not necessarily of constant size**	
		$UCL_{p_i} = \bar{p} + 3\frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n_i}}$	$LCL_{p_i} = \bar{p} - 3\frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n_i}}$
		If the Sample size is constant (n)	
		$UCL_p = \bar{p} + 3\frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}}$	$LCL_p = \bar{p} - 3\frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}}$





p chart- example

Data for defective CDs from 20 samples (sample size = 100) are shown in table below:

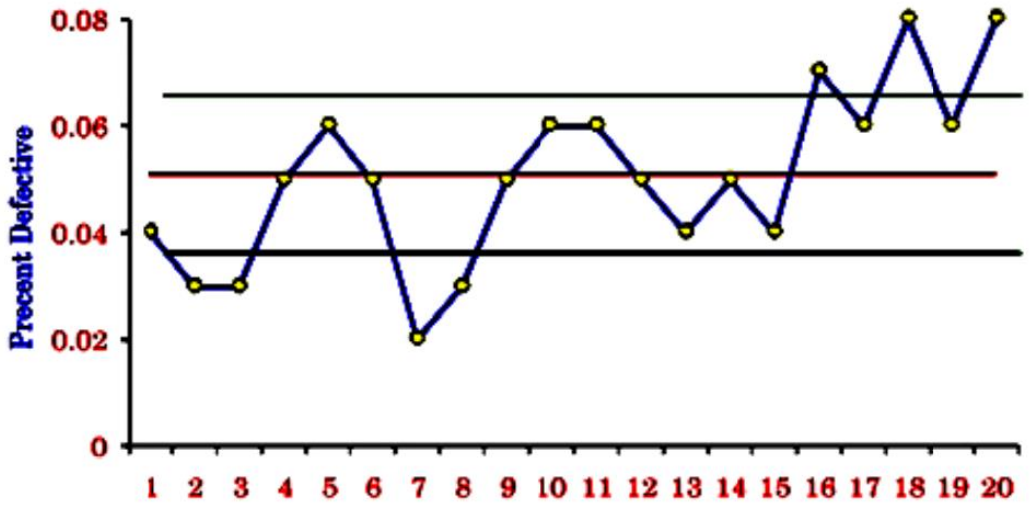
Sample No.	No. of Defective CDs = x	Proportion Defective = x/sample size	Sample No.	No. of Defective CDs = x	Proportion Defective = x/sample size
1	4	0.04	11	6	0.06
2	3	0.03	12	5	0.05
3	3	0.03	13	4	0.04
4	5	0.05	14	5	0.05
5	6	0.06	15	4	0.04
6	5	0.05	16	7	0.07
7	2	0.02	17	6	0.06
8	3	0.03	18	8	0.08
9	5	0.05	19	6	0.06
10	6	0.06	20	8	0.08

p chart- example

$$CL = \frac{\text{Sum of defectives}}{\text{Sum of all samples}} = \frac{101}{2000} = 0.051$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.051 + 3\sqrt{\frac{0.051(1-0.051)}{100}} = 0.066$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{N}} = 0.051 - 3\sqrt{\frac{0.051(1-0.051)}{100}} = 0.036$$



np chart

np chart for number / rate of units in a category	$CL_{np} = \overline{np}$	$UCL_{np} = \overline{np} + 3\sqrt{\overline{np}(1-\bar{p})}$	$LCL_{np} = \overline{np} - 3\sqrt{\overline{np}(1-\bar{p})}$





np chart example

Smartbulbs Inc is a famous LED bulb manufacturer. The supervisor drew a randomly constant sample size of 200 bulbs every hour and reported the number of defective bulbs for each lot. Based on the given data, prepare the control chart for the number of defectives and determine whether the process is in statistical control.

Lot	Sample Size	Number of defective in the sample (np)
1	200	4
2	200	8
3	200	6
4	200	6
5	200	4
6	200	8
7	200	2
8	200	1
9	200	9
10	200	6
11	200	8
12	200	1
13	200	2
14	200	9
15	200	4
16	200	3
17	200	9
18	200	6
19	200	2
20	200	7

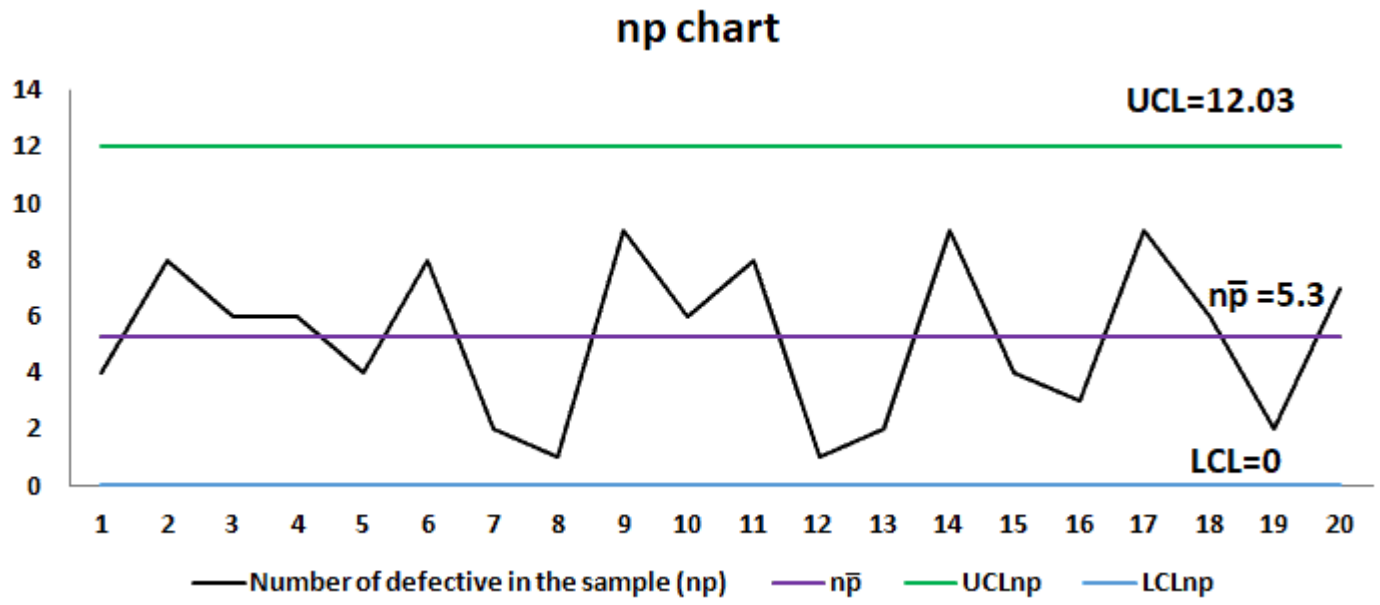




np chart example

$$UCL_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})} = 5.3 + 3\sqrt{5.3 * 0.9738} = 12.03$$

$$LCL_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})} = 5.3 - 3\sqrt{5.3 * 0.9738} = -1.5 = 0$$



c chart

$$\bar{c} = \frac{\text{Sum of defects in all samples}}{\text{Total number of samples}} = \text{Centre line of } c \text{ chart}$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} \qquad \text{and} \qquad LCL = \bar{c} - 3\sqrt{\bar{c}}$$





c chart example

Data for defects on TV set from 20 samples (sample size = 10) are shown in the table below:

Sample No.	No. of Defects	Sample No.	No. of Defects	Sample No.	No. of Defects	Sample No.	No. of Defects
1	5	6	4	11	6	16	5
2	4	7	5	12	5	17	4
3	5	8	6	13	4	18	6
4	6	9	8	14	7	19	6
5	4	10	7	15	6	20	6





c chart example

$$CL = \frac{\text{Sum of defects}}{\text{Number of samples}}$$

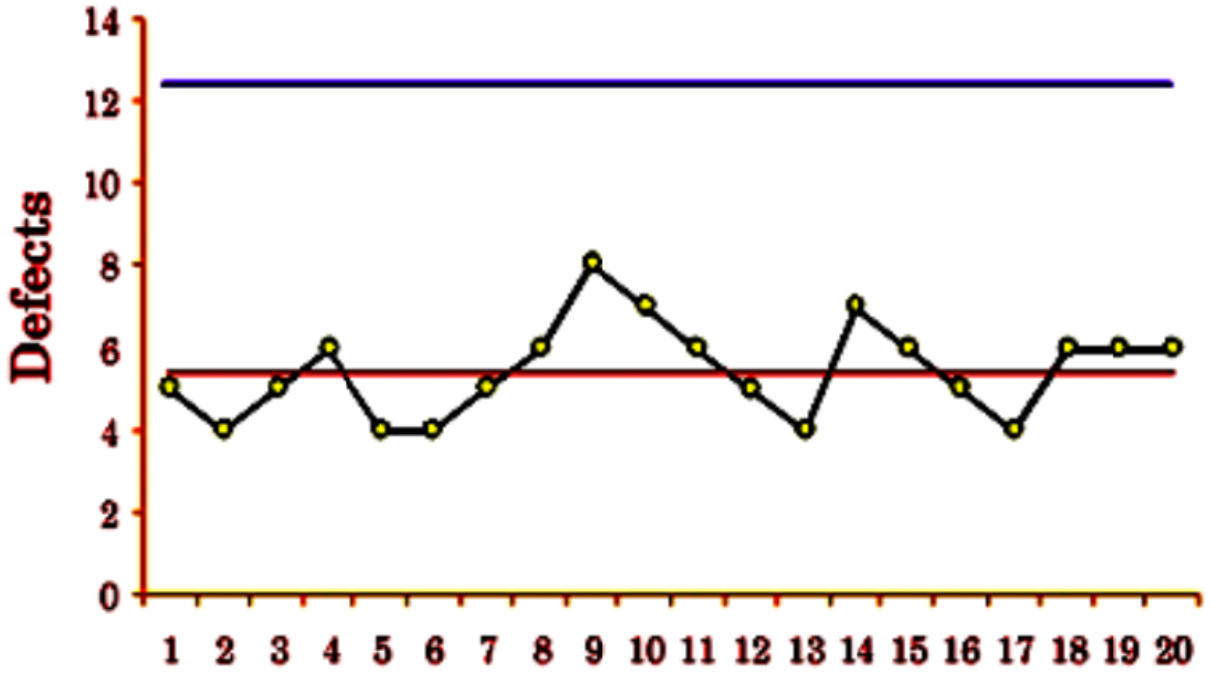
$$= \frac{109}{20} = 5.45$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 5.45 + 3\sqrt{5.45}$$

$$= 12.45$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 5.45 - 3\sqrt{5.45}$$

$$= -1.55 = 0$$



u chart

$$\bar{u} = \frac{\text{Sum of defects in all samples}}{\text{Total number of items in all samples}} = \frac{\sum_{i=1}^{k} c_i}{\sum_{i=1}^{k} n_i}$$

c_i = Number of defects in i^{th} sample

k = Number of samples

n_i = Size of ith samples

$$UCL = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}} \quad \text{and} \quad LCL = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}}$$





u chart

u chart for number of incidences per unit in one or more categories	$CL_u = \bar{u}$	Samples not necessarily of constant size	
		$UCL_u = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$	$LCL_u = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}}$
		using average sample size	
		$UCL_u = \bar{u} + 3\sqrt{\frac{\bar{u}}{\bar{n}}}$	$UCL_u = \bar{u} - 3\sqrt{\frac{\bar{u}}{\bar{n}}}$
		If the sample size is constant (n)	
		$UCL_u = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}}$	$UCL_u = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$



u chart example

The following data shows the number of defects per lot in 20 consecutive lots of 100 motorbikes each. Plot the average number of defects per unit based on the data and determine the process in statistical control.

Lot	Sample Size	Number of defects in the sample (c)
1	100	52
2	100	48
3	100	56
4	100	42
5	100	39
6	100	39
7	100	54
8	100	56
9	100	41
10	100	43
11	100	47
12	100	52
13	100	44
14	100	47
15	100	50
16	100	40
17	100	47
18	100	46
19	100	44
20	100	50





Lot	Sample Size	Number of defects in the sample (c)	u =c/n
1	100	52	0.520
2	100	48	0.480
3	100	56	0.560
4	100	42	0.420
5	100	39	0.390
6	100	39	0.390
7	100	54	0.540
8	100	56	0.560
9	100	41	0.410
10	100	43	0.430
11	100	47	0.470
12	100	52	0.520
13	100	44	0.440
14	100	47	0.470
15	100	50	0.500
16	100	40	0.400
17	100	47	0.470
18	100	46	0.460
19	100	44	0.440
20	100	50	0.500

u chart example

A motorbike manufacturer collected the number of defects identified in the final inspection stage for 20 days. Based on the given data, the supervisor wants to know the quality level using a control chart for defects per motorbike and determine the process in statistical control.

Lot	Sample Size	Number of defects in the sample (c)
1	100	52
2	120	48
3	150	56
4	80	25
5	130	39
6	125	39
7	140	54
8	90	70
9	100	41
10	110	43
11	115	47
12	120	52
13	90	44
14	85	47
15	100	50
16	120	40
17	115	47
18	120	46
19	120	44
20	110	50

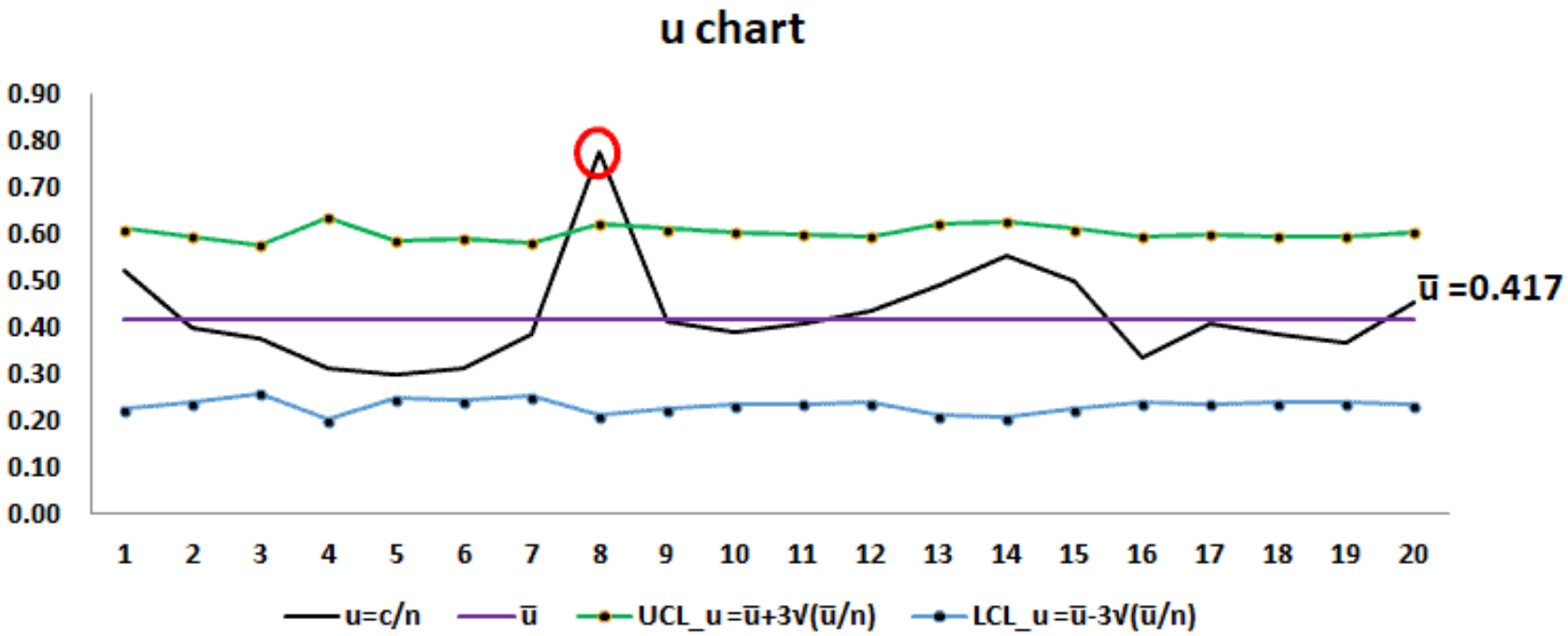
u chart example

Lot	Sample Size	Number of defects in the sample (c)	u =c/n	$\bar{u}$	UCLu	LCLu
1	100	52	0.52000	0.4170	0.61068	0.22325
2	120	48	0.40000	0.4170	0.59380	0.24012
3	150	56	0.37333	0.4170	0.57513	0.25879
4	80	25	0.31250	0.4170	0.63355	0.20038
5	130	39	0.30000	0.4170	0.58687	0.24706
6	125	39	0.31200	0.4170	0.59023	0.24370
7	140	54	0.38571	0.4170	0.58069	0.25324
8	90	70	0.77778	0.4170	0.62116	0.21277
9	100	41	0.41000	0.4170	0.61068	0.22325
10	110	43	0.39091	0.4170	0.60167	0.23226
11	115	47	0.40870	0.4170	0.59761	0.23632
12	120	52	0.43333	0.4170	0.59380	0.24012
13	90	44	0.48889	0.4170	0.62116	0.21277
14	85	47	0.55294	0.4170	0.62708	0.20685
15	100	50	0.50000	0.4170	0.61068	0.22325
16	120	40	0.33333	0.4170	0.59380	0.24012
17	115	47	0.40870	0.4170	0.59761	0.23632
18	120	46	0.38333	0.4170	0.59380	0.24012
19	120	44	0.36667	0.4170	0.59380	0.24012
20	110	50	0.45455	0.4170	0.60167	0.23226





u chart example



SIX SIGMA (6σ)



Lesser number of cars



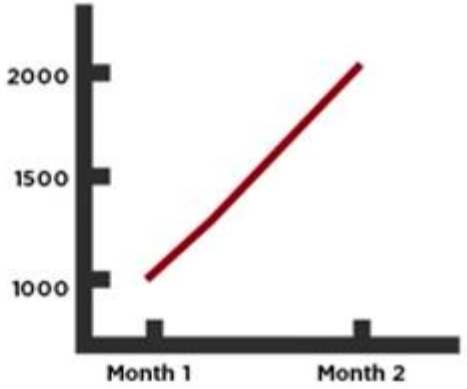


SIX SIGMA (6σ)

The question you are faced with:



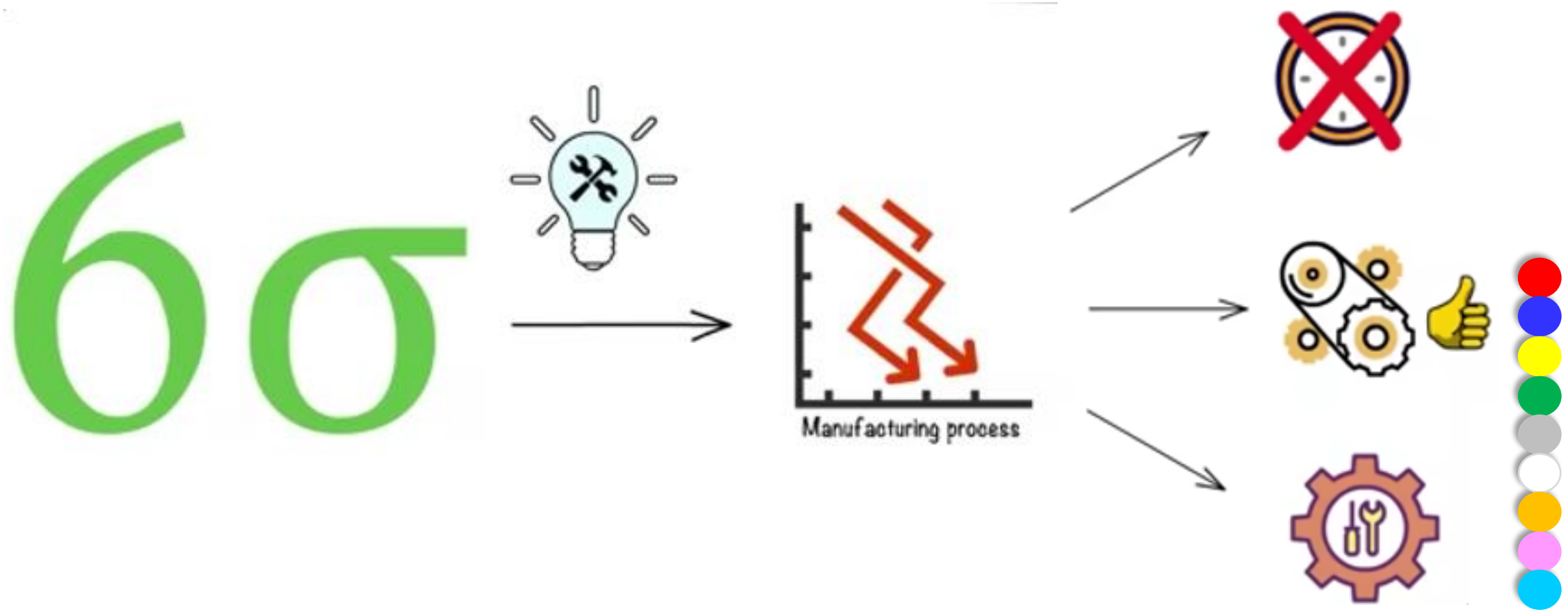
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SIX SIGMA (6σ)



SIX SIGMA (6σ)

Six Sigma aims to reduce



Time

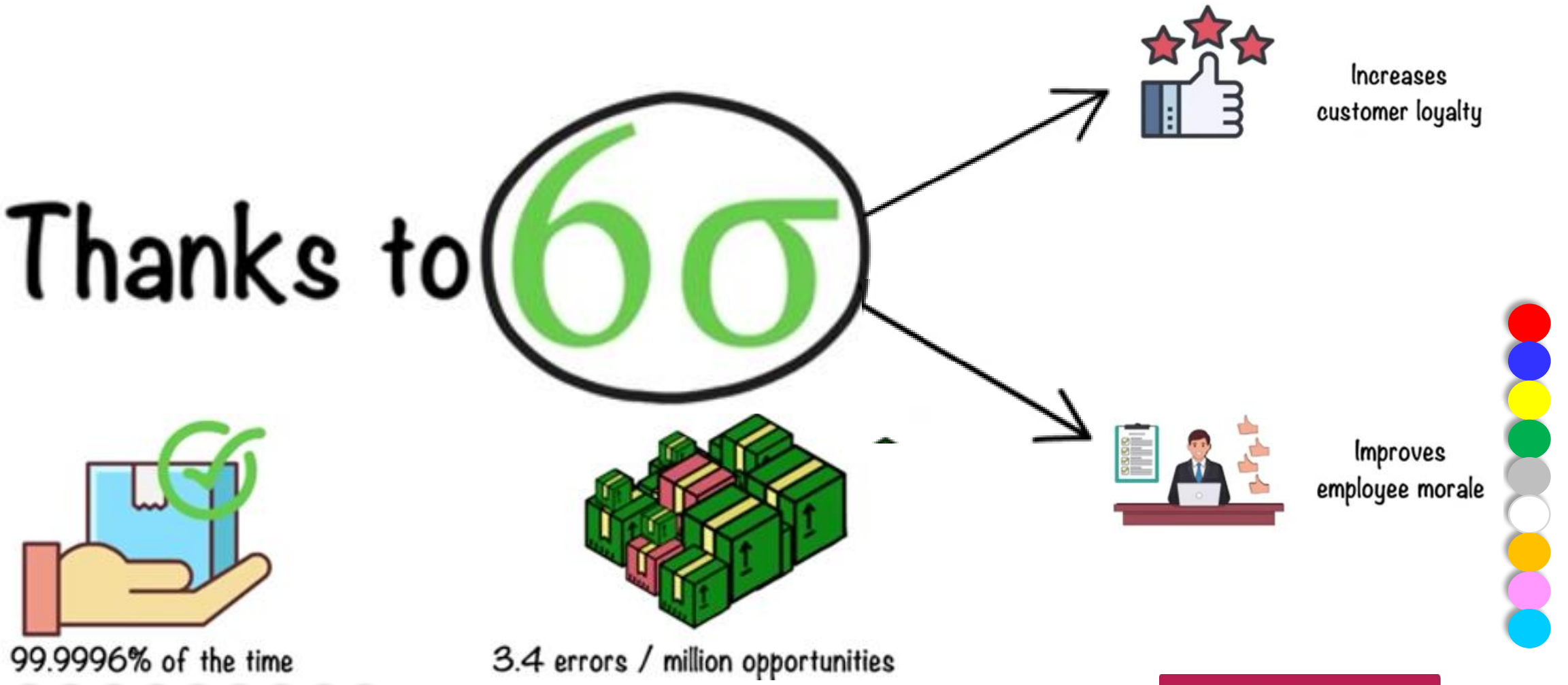


Defects

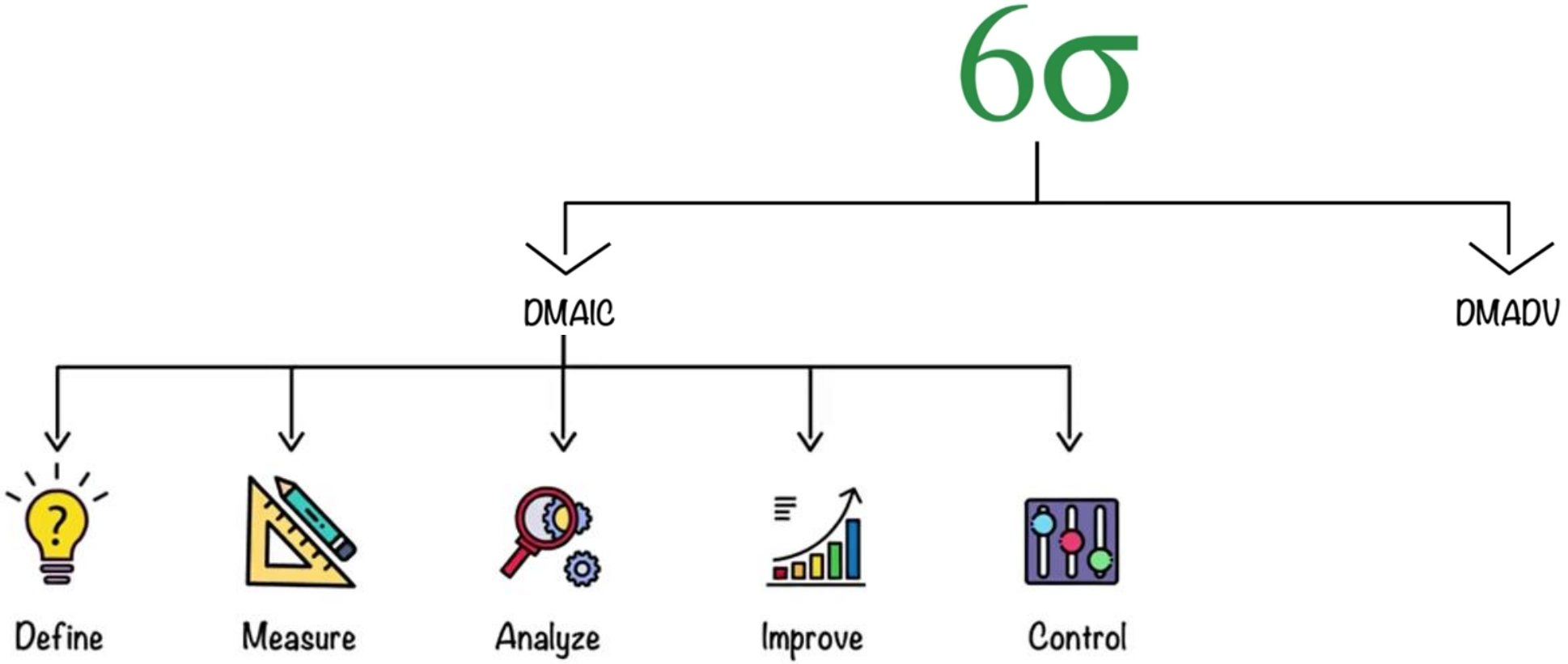


Variability

SIX SIGMA (6σ)


Thanks to 6σ
Increases customer loyalty
Improves employee morale
99.9996% of the time
3.4 errors / million opportunities

SIX SIGMA (6σ)



SIX SIGMA (6σ)



Define



Determine what issues you are facing



Opportunities for improvement



Customer requirements





&







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SIX SIGMA (6σ)



Measure



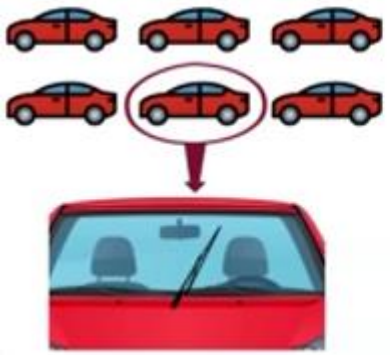
Process

(Unaltered)



Manufactured in a day

In the current scenario



1000 cars are manufactured in a day

By 1/30 machines

Some of the metrics measured are



- [x] Number of cars produced in a day
- [x] Time taken to assemble a car
- [ ] Number of windshield wipers attached in a day
- [ ] Time taken for attaching windshield wipers
- [ ] Number of defects detected from each machine

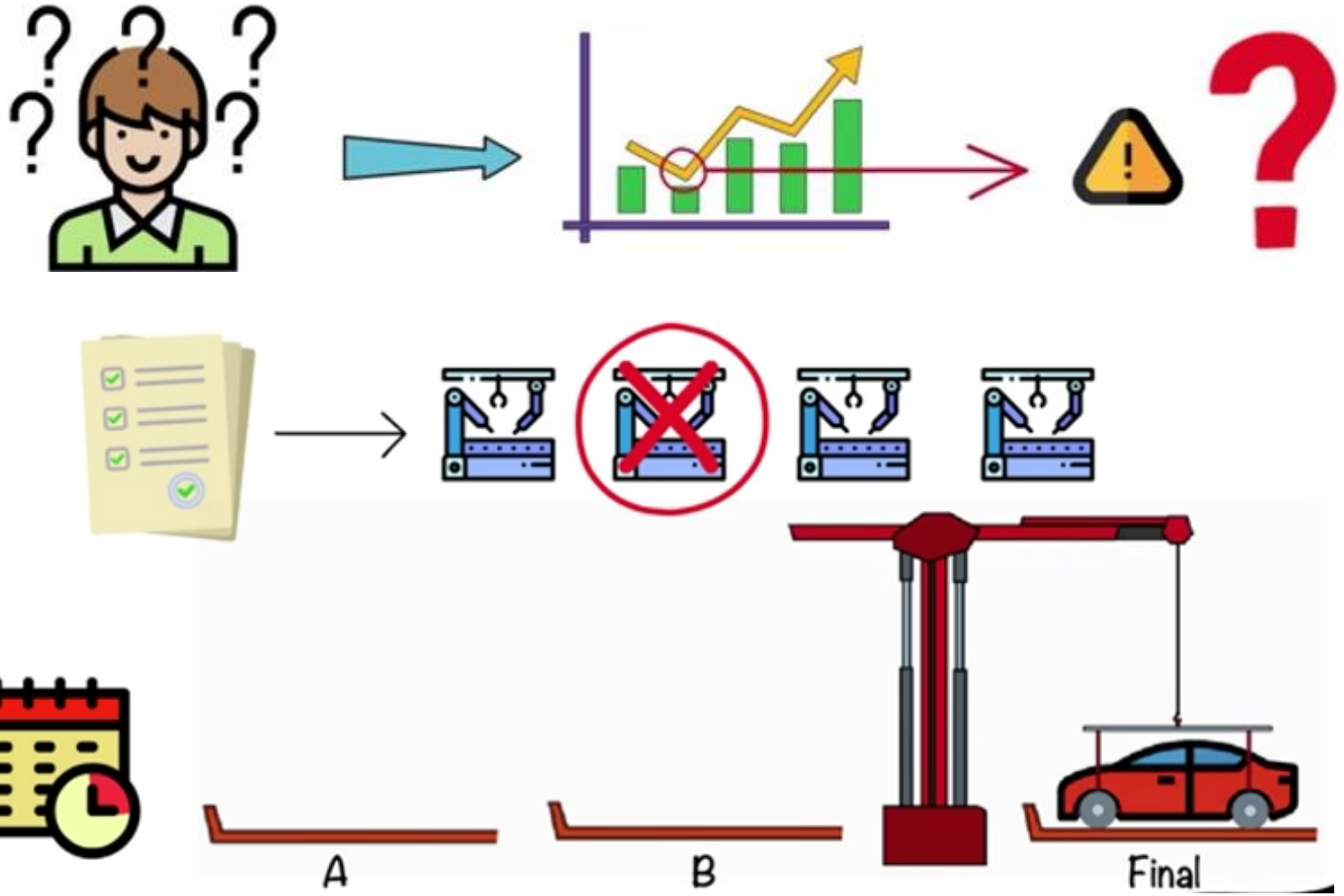




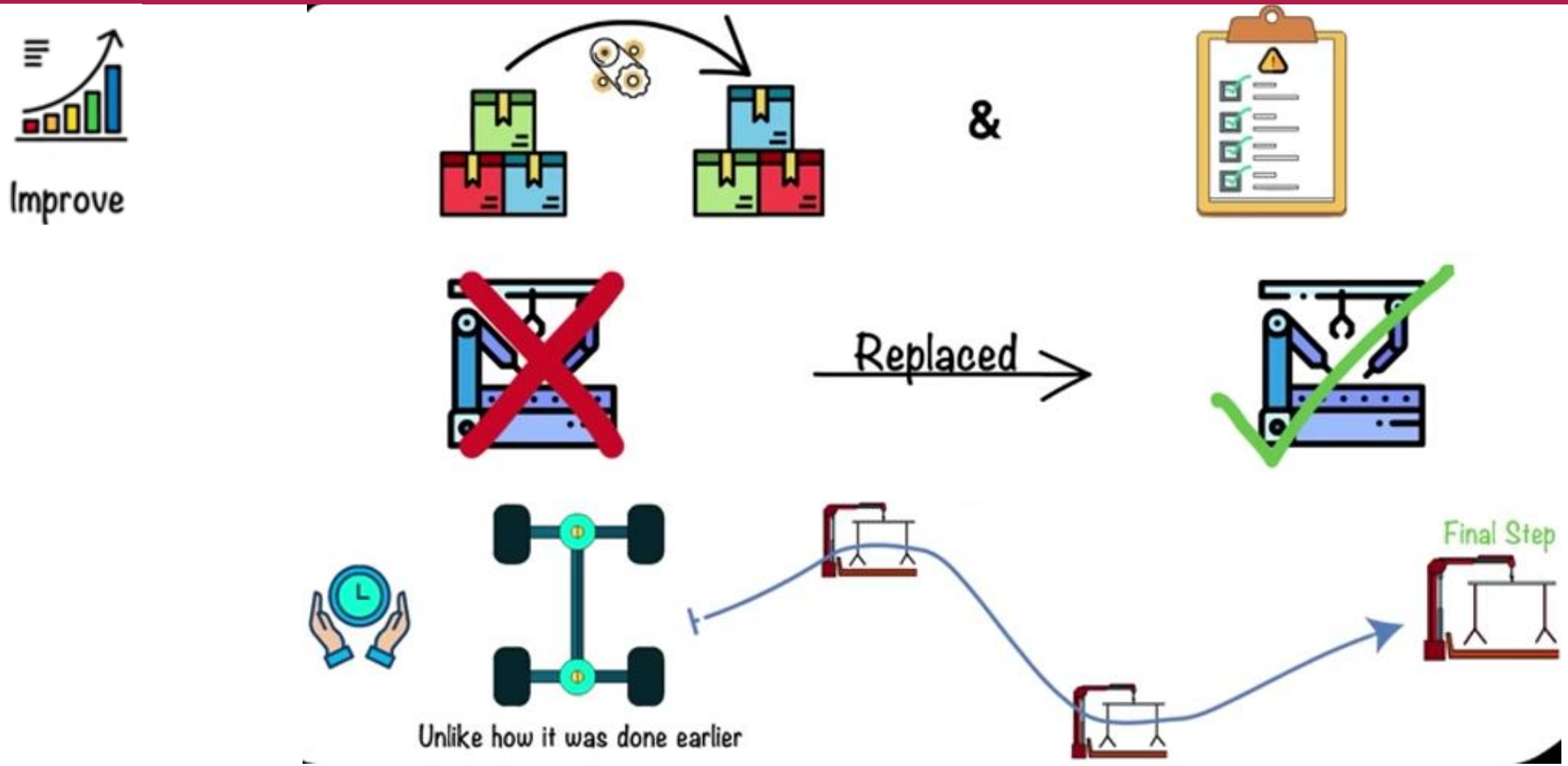
SIX SIGMA (6σ)



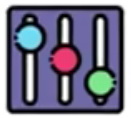
Analyze



SIX SIGMA (6σ)

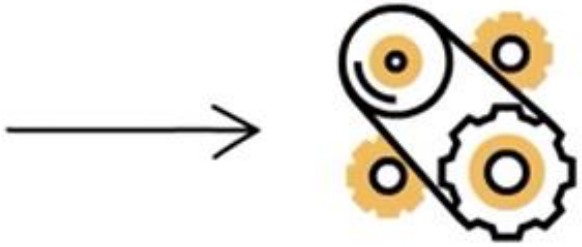


SIX SIGMA (6σ)



Control



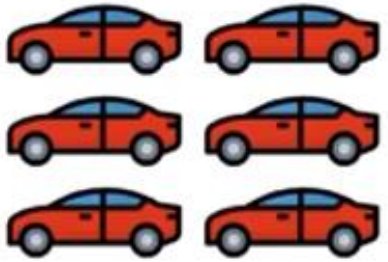


&



Based on the changes made





1,000

2,000

cars a day

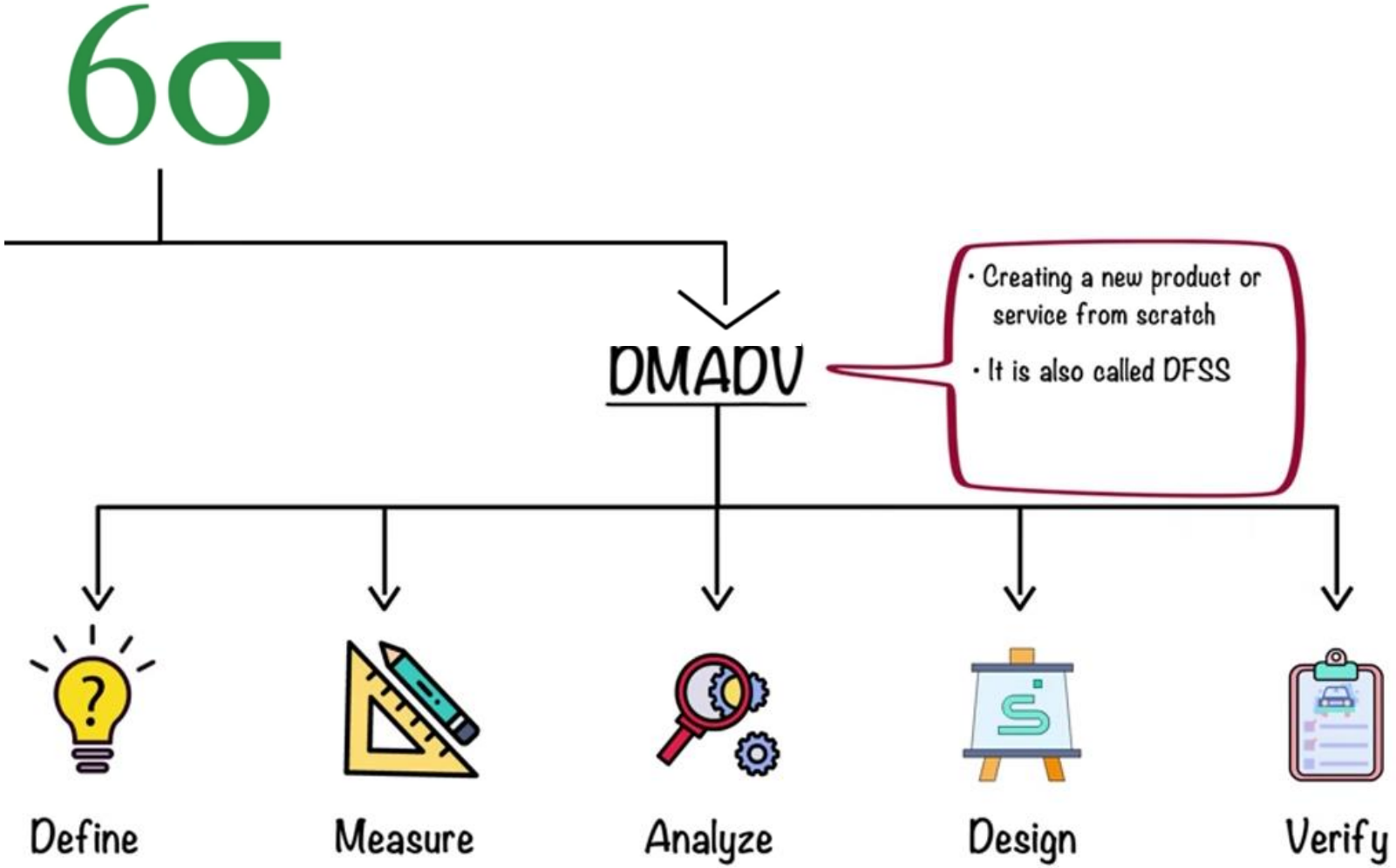






DMAIC is the most commonly used methodology in the world

SIX SIGMA (6σ)


6σ
DMADV
• Creating a new product or service from scratch
• It is also called DFSS
Define
Measure
Analyze
Design
Verify

SIX SIGMA (6σ)



Define



Define requirement of the customer

Input from customer





Historical data

Industrial research









Collected data



150 mph

Customers are also more incline towards cars which have



&







SIX SIGMA (6σ)



Measure

Use customer requirement



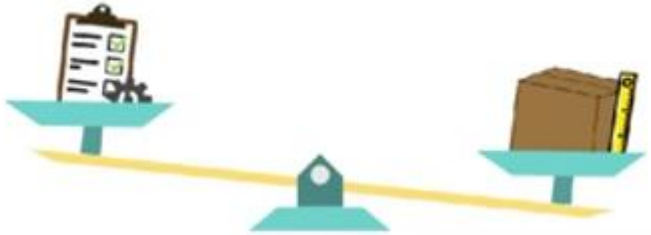
Create specifications



Help to find in a measurable product



Collect and measure the data



Some of the major specifications that you focus on are



Top speed



Engine type

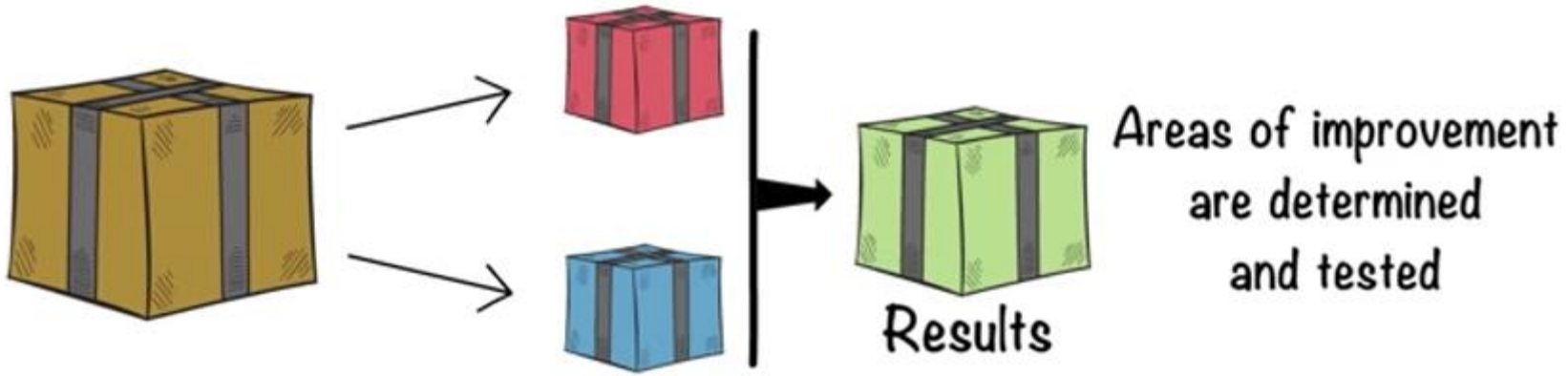


Type of frame

SIX SIGMA (6σ)



Analyze





Create prototype

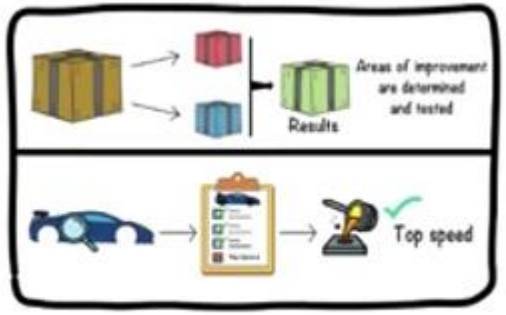
Observation acceptable (except TOP SPEED)

Research on alternative materials (say Al alloy)

SIX SIGMA (6σ)



Design



Analyze phase

Based on the analysis phase, a new process or product is designed



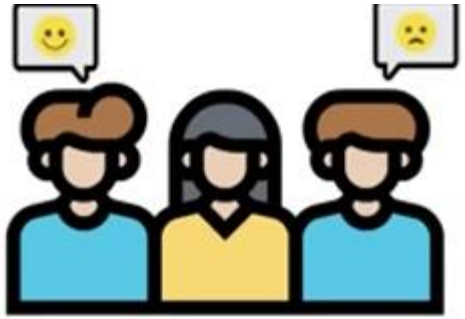
or







Manufacturing with new material





Again analyze



Further changes are made



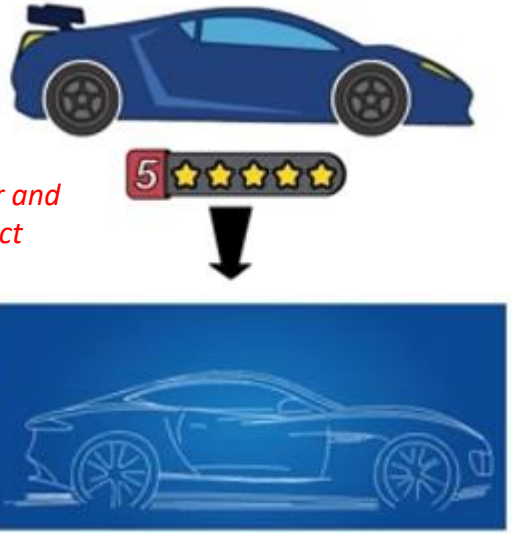


SIX SIGMA (6σ)



Verify





Get feedback from customer and incorporate in new product

And guess what?



SIX SIGMA (6σ)

Six sigma has also found success in a number of different industries



Petrochemical



Healthcare



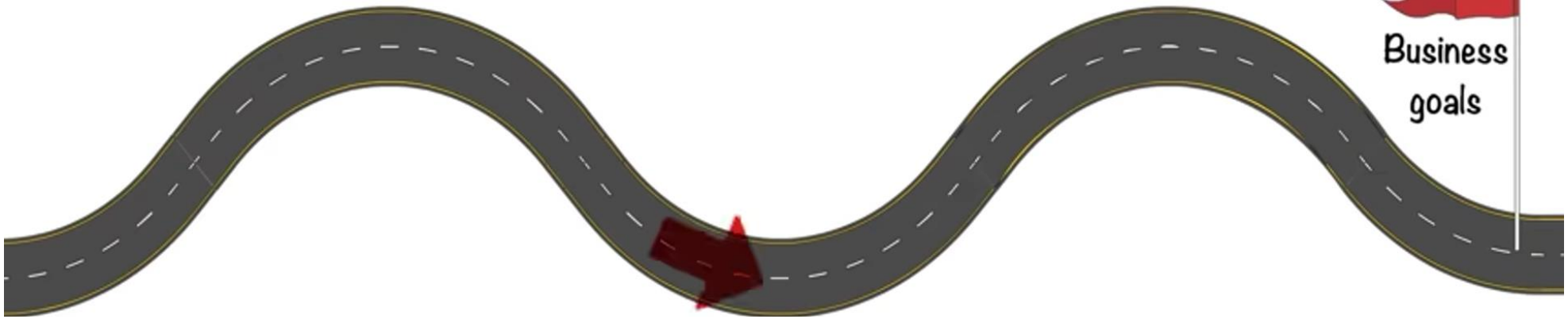
Banking



Government



Software



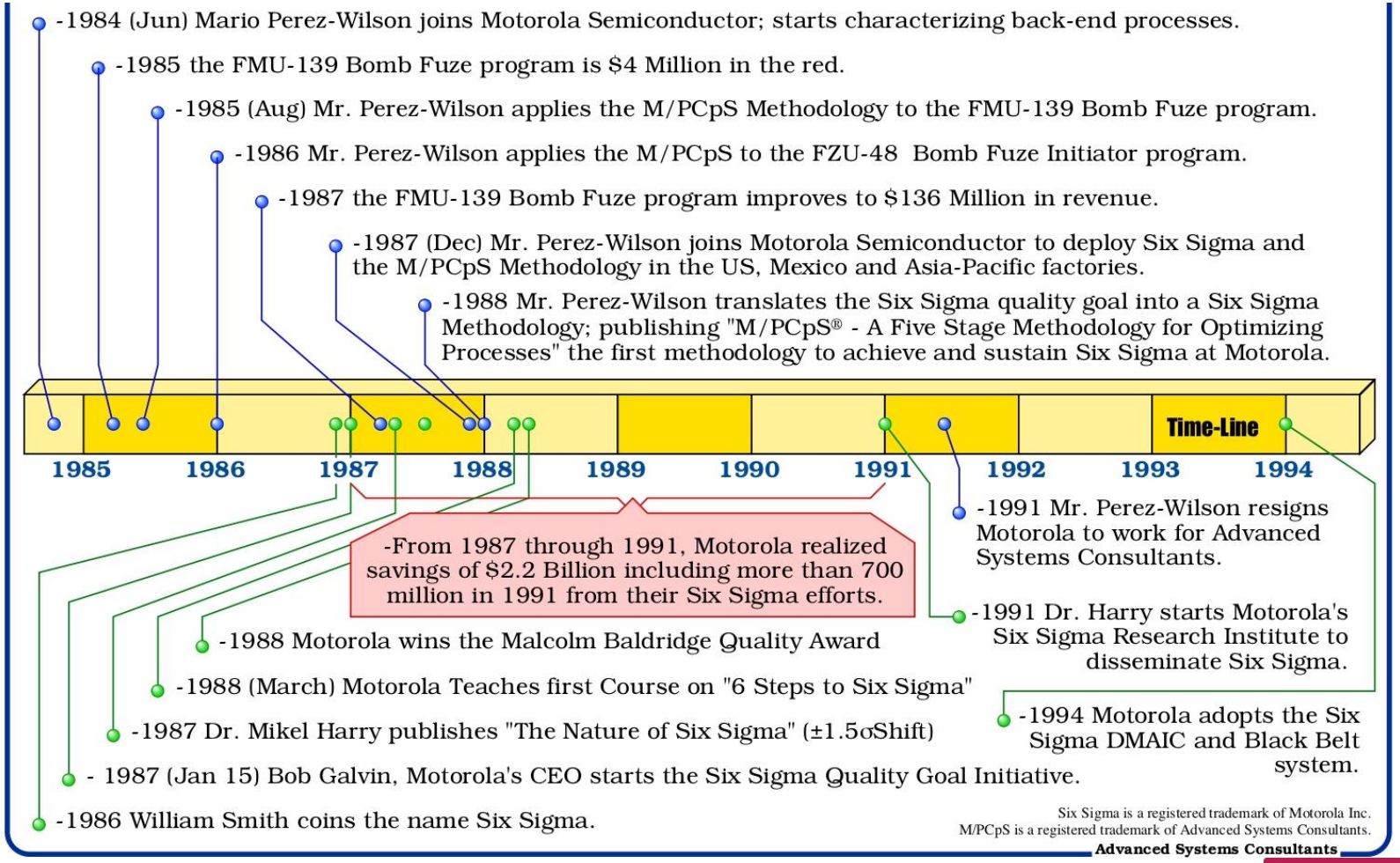
SIX SIGMA (6σ)

- Six Sigma is a disciplined, statistical-based, data-driven approach and continuous improvement methodology for eliminating defects in a product, process or service.
- Six Sigma is a methodology used to improve business processes by utilizing statistical analysis rather than guesswork.
- Processes are improved by controlling variation and understanding the intricacies within them.
- Results in more predictable and profitable business processes.
- Six Sigma is more than “training”; it is an approach based on data and geared toward projects with quantifiable business outcomes.
- Six Sigma quality is achieved when long-term defect levels are below 3.4 defects per million opportunities (DPMO)



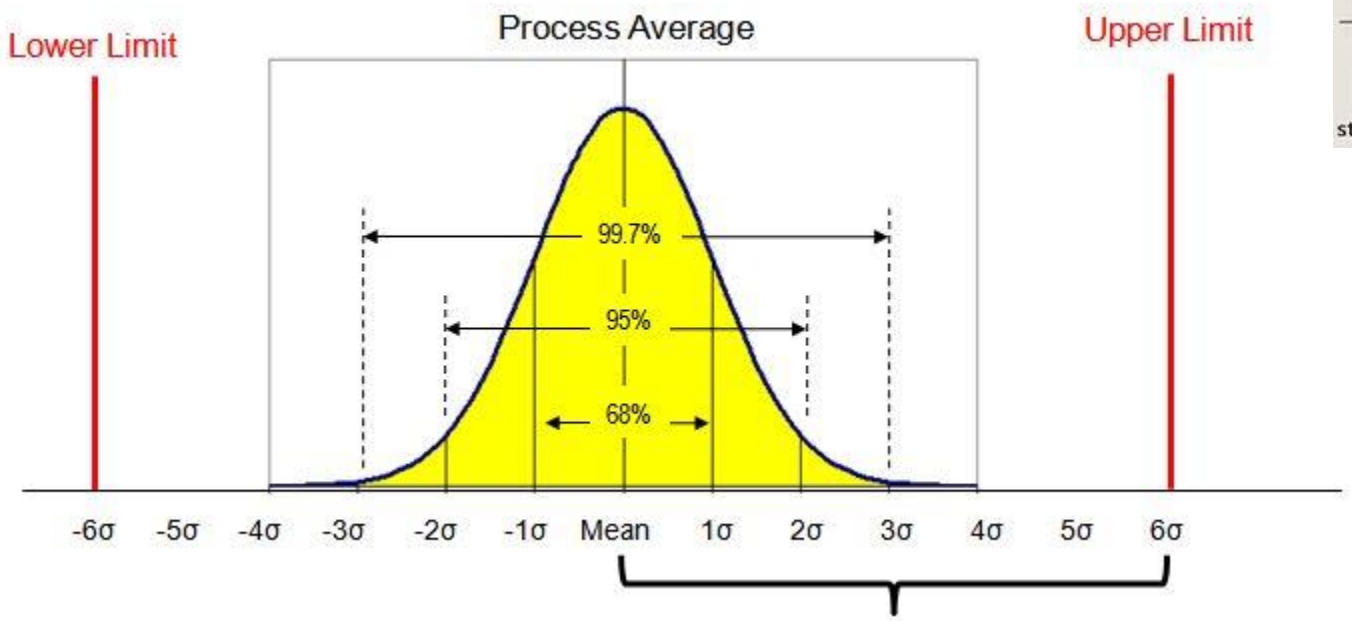


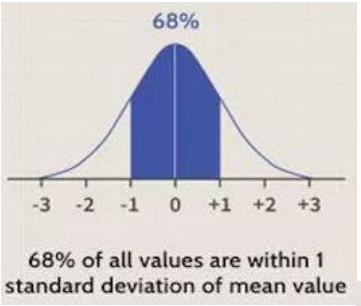
6σ timeline

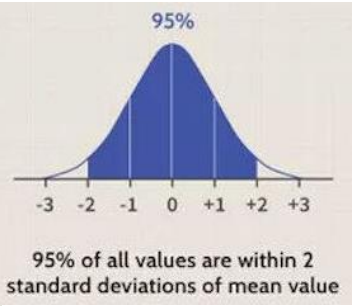

-1984 (Jun) Mario Perez-Wilson joins Motorola Semiconductor; starts characterizing back-end processes.
-1985 the FMU-139 Bomb Fuze program is $4 Million in the red.
-1985 (Aug) Mr. Perez-Wilson applies the M/PCpS Methodology to the FMU-139 Bomb Fuze program.
-1986 Mr. Perez-Wilson applies the M/PCpS to the FZU-48 Bomb Fuze Initiator program.
-1987 the FMU-139 Bomb Fuze program improves to $136 Million in revenue.
-1987 (Dec) Mr. Perez-Wilson joins Motorola Semiconductor to deploy Six Sigma and the M/PCpS Methodology in the US, Mexico and Asia-Pacific factories.
-1988 Mr. Perez-Wilson translates the Six Sigma quality goal into a Six Sigma Methodology; publishing "M/PCpS® - A Five Stage Methodology for Optimizing Processes" the first methodology to achieve and sustain Six Sigma at Motorola.
Time-Line
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
-1991 Mr. Perez-Wilson resigns Motorola to work for Advanced Systems Consultants.
-From 1987 through 1991, Motorola realized savings of $2.2 Billion including more than 700 million in 1991 from their Six Sigma efforts.
-1991 Dr. Harry starts Motorola's Six Sigma Research Institute to disseminate Six Sigma.
-1988 Motorola wins the Malcolm Baldridge Quality Award
-1988 (March) Motorola Teaches first Course on "6 Steps to Six Sigma"
-1994 Motorola adopts the Six Sigma DMAIC and Black Belt system.
-1987 Dr. Mikel Harry publishes "The Nature of Six Sigma" (±1.5σShift)
- 1987 (Jan 15) Bob Galvin, Motorola's CEO starts the Six Sigma Quality Goal Initiative.
-1986 William Smith coins the name Six Sigma.
Six Sigma is a registered trademark of Motorola Inc.
M/PCpS is a registered trademark of Advanced Systems Consultants.
Advanced Systems Consultants

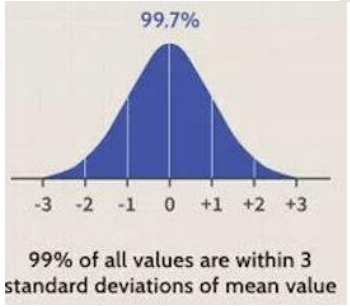
6σ Bell curve

- The term “Six Sigma” is derived from the **bell curve** in statistics, in which sigma represents the standard deviation from the center.









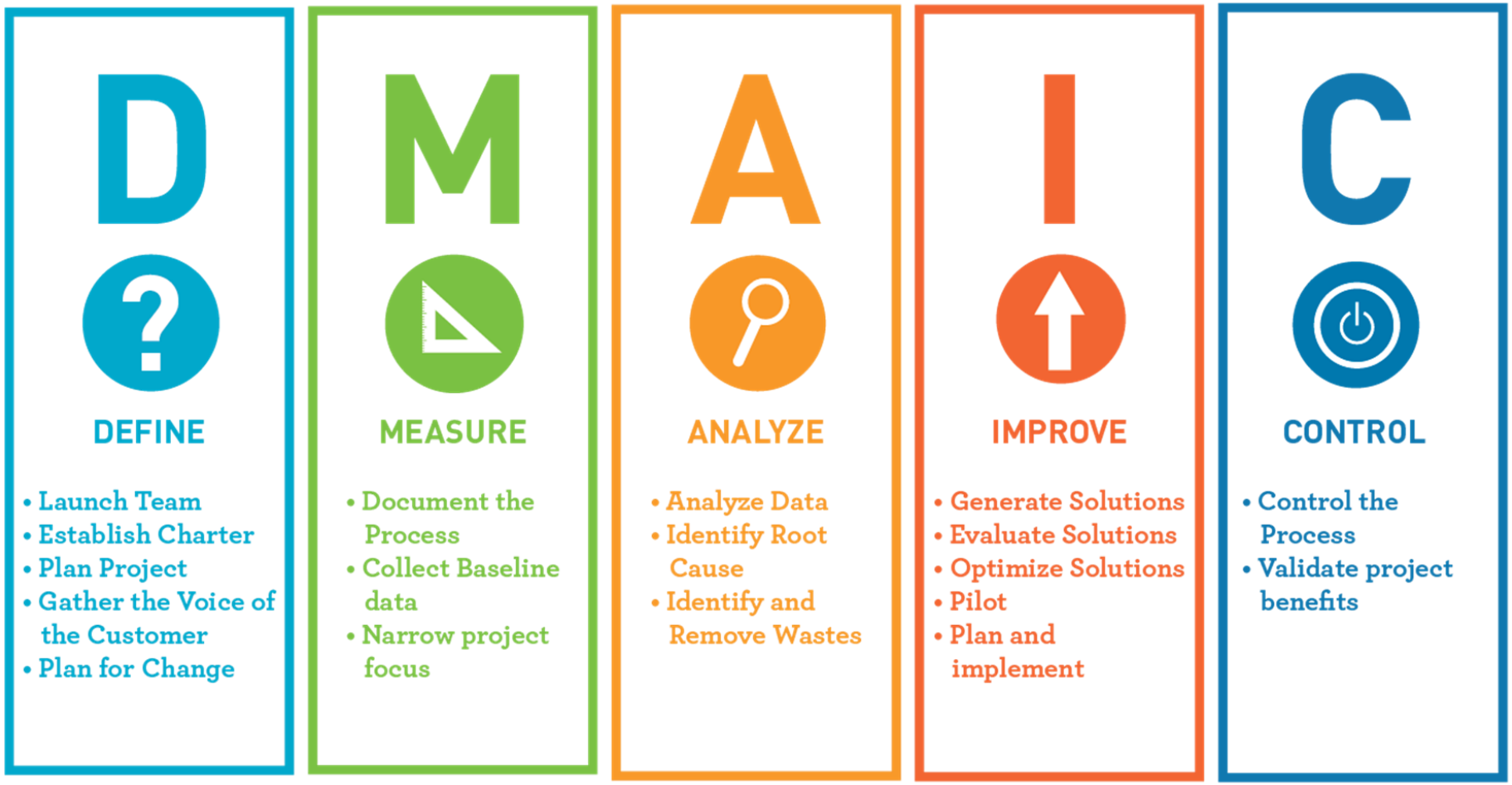




Objectives


Improve customer satisfaction
Reduce cycle time
Reduce errors
Improve quality
Eliminate defects
Objectives of Six Sigma

6σ METHODS - DMAIC



DMAIC

The DMAIC method is used primarily for improving existing business processes.

- **D**efine the problem and the project goals
- **M**easure in detail the various aspects of the current process
- **A**nalyze data to, among other things, find the root defects in a process
- **I**mprove the process
- **C**ontrol how the process is done in the future

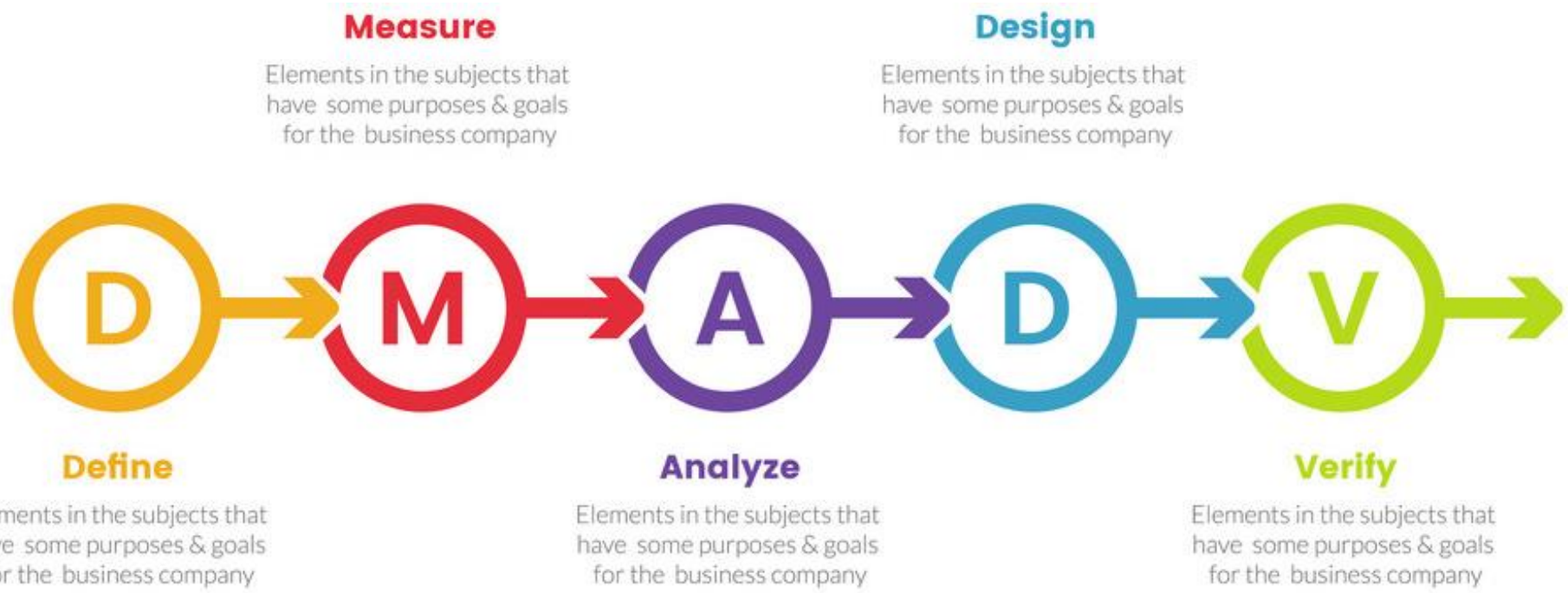




Tools used for DMAIC

D	M	A	I	C
• Project Charter • Multi-Generational Project Planning • SIPOC • Risk Management Plan • Stakeholder Analysis • Communication Plan • Project Plan • VOC Plan • Affinity Diagram • Structure Tree Diagram • Interviews • Surveys	• Value Stream Map • Detailed Process Map • Input – Output Map • Spaghetti Diagram • C&E Matrix • Data Collection Plan • Basic Statistical Tools • Graphical Tools • Pareto • Measurement System Analysis • Control Charts • Process Capability	• Fishbone Diagram • 5 Why's • FMEA • Multi Vari Analysis • Means, Medians and Variance Tests • Proportion Tests • Chi Square • Regression • Non Parametric Tests • Non Value Added Analysis • Takt Time • Little's Law • Process Value Efficiency • Constraint ID • Line Balancing	• Full and Fractional Factorial DOE • Brainstorming • Force Field Analysis • Process Flow Improvement • 5S • Replenishment Pull • Cellular Layout • Setup Reduction • CONWIP • Batch Size Optimization • Total Productive Maintenance • ABC Stratification • Future State Process Map • Solution Matrix • Implementation Plan	• Implementation Checklist • Process FMEA • Process Control and Audit Plan • Visual Process Controls • Control Charts • Mistake Proofing • Standard Work

6σ METHODS - DMADV



DMADV

The DMADV method is typically used to create new processes and new products or services.

- **D**efine the project goals
- **M**easure critical components of the process and the product capabilities
- **A**nalyze the data and develop various designs for the process, eventually picking the best one
- **D**esign and test details of the process
- **V**erify the design by running simulations and a pilot program, and then handing over the process to the client





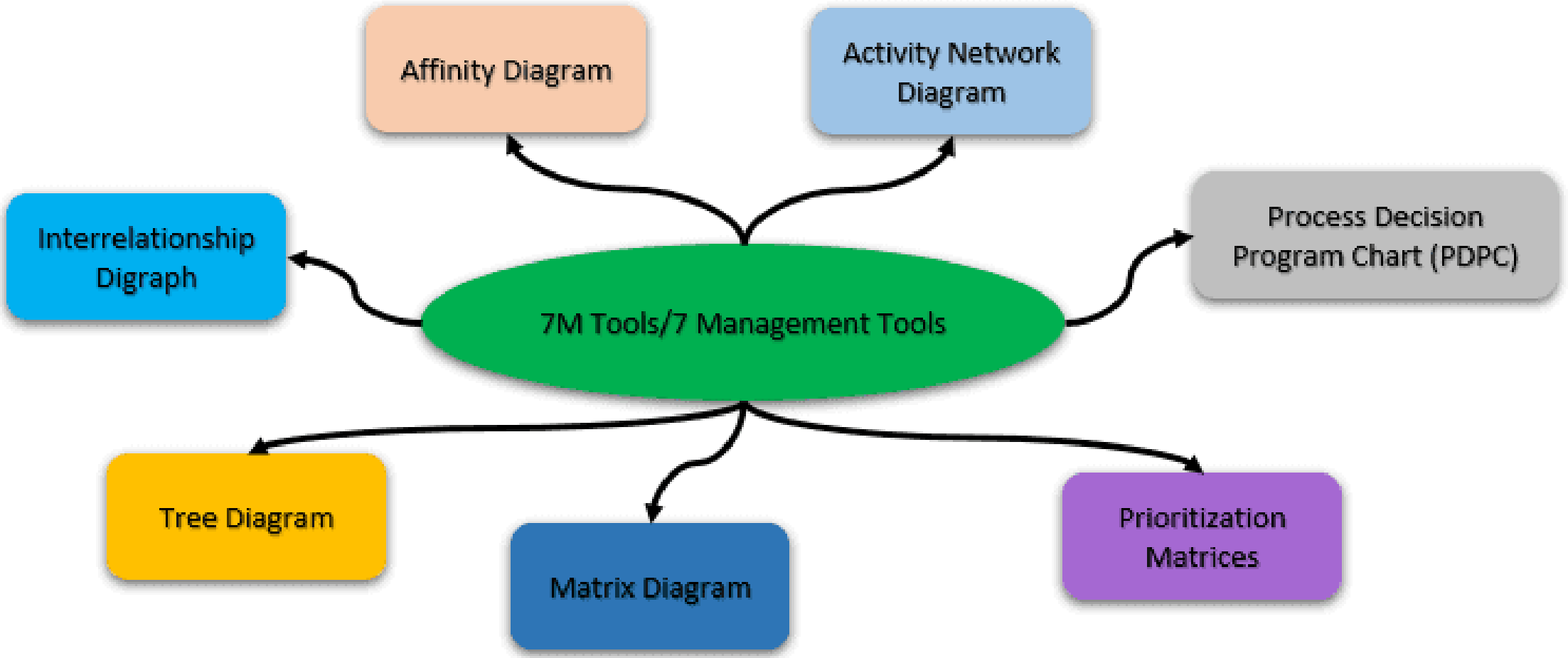
Six sigma – training



Benefits of 6σ

- Increased productivity, efficiency, and effectiveness
- Improved quality of product and services and a reduction in waste
- Reduced operating costs and process variation
- Higher customer satisfaction and competitive advantage
- Improved communication and employee engagement

NEW 7 MANAGEMENT TOOLS



NEW 7 MANAGEMENT TOOLS

1. **Affinity diagram** organize large amounts of poorly organized or unorganized data into groups that reflect natural relationships

2. **Activity network diagram (or) Arrow diagram**: Shows the required order of tasks in a project or process, the best schedule for the entire project, and potential scheduling and resource problems and their solutions.

3. **Process decision Program chart**: Systematically identifies what might go wrong in a plan under development.

4. **Prioritization matrices**: Shows the relationship between two, three, or four groups of information and can give information about the relationship, such as its strength, the roles played by various individuals, or measurements.

5. **Matrix data analysis**: A complex mathematical technique for analyzing matrices, often replaced by a similar prioritization matrix. A prioritization matrix is an L-shaped matrix that uses pairwise comparisons of a list of options to a set of criteria to choose the best option(s).

6. **Tree diagram**: Breaks down broad categories into finer and finer levels of detail, helping to move step-by-step thinking from generalities to specifics.

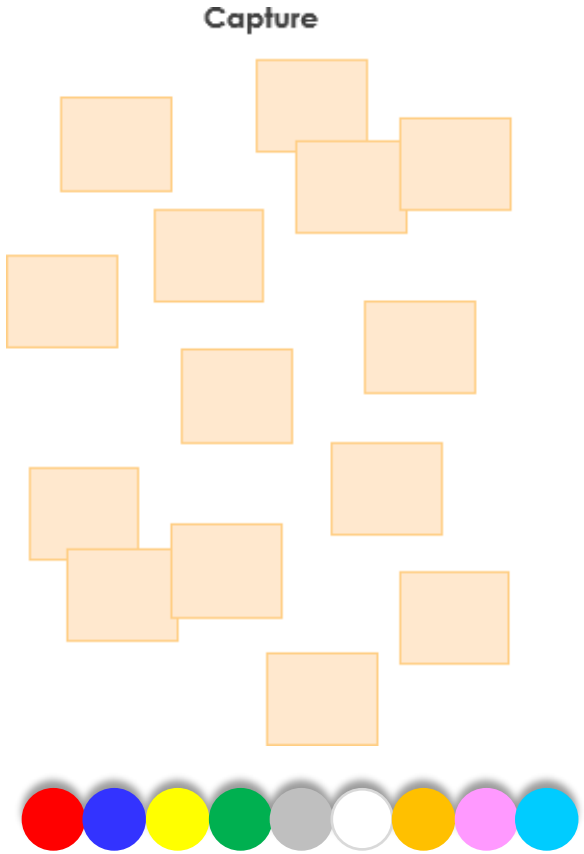
7. **Interrelationship diagram**: Shows cause-and-effect relationships and helps analyze the natural links between different aspects of a complex situation.

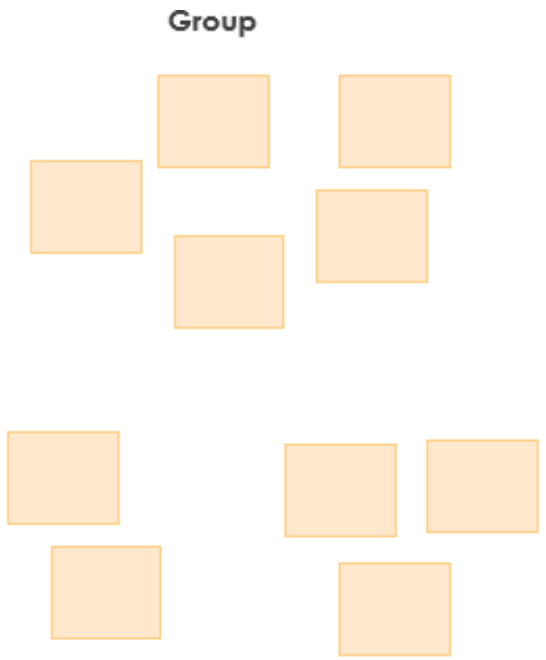


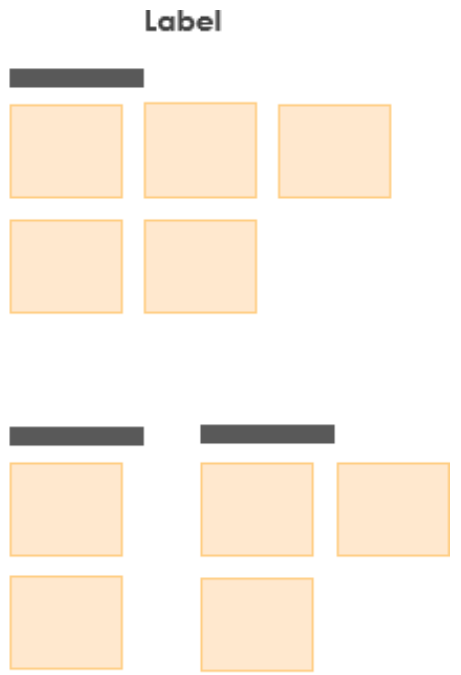


Affinity Diagram

1. **Capture**: Record each idea with a marking pen on a separate sticky note or card
2. **Group**: Look for ideas that seem to be related in some way and place them side by side
3. **Label**: labeling each group of cards into categories and eliminate duplicate ideas.



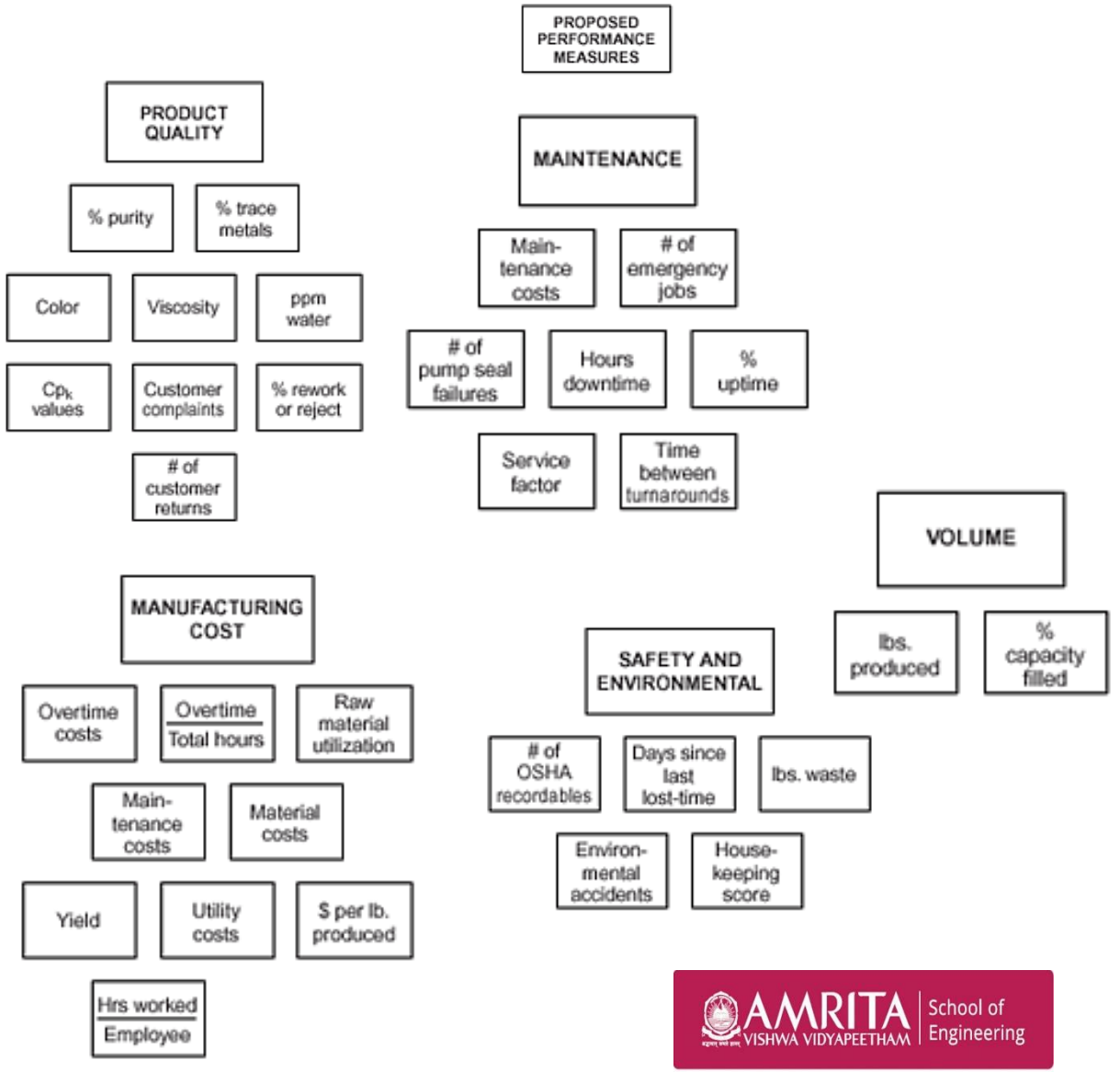






Affinity Diagram example

Possible Performance Measures	
% purity	# of OSHA recordables
% trace metals	# of customer returns
Maintenance costs	Customer complaints
# of emergency jobs	Overtime/total hours worked
lbs. produced	$/lb. produced
Environmental accidents	Raw material utilization
Material costs	Yield
Overtime costs	Utility cost
# of pump seal failures	ppm water
Viscosity	Color
Cp_k values	Service factor
Safety	Time between turnarounds
Days since last lost-time	Hours worked/employee
% rework or reject	lbs. waste
Hours downtime	Housekeeping score
% uptime	% capacity filled



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When to use affinity diagram?

- When confronted with many facts or ideas in apparent chaos
- When issues seem too large and complex to grasp
- When group consensus is necessary

Typical situations are:

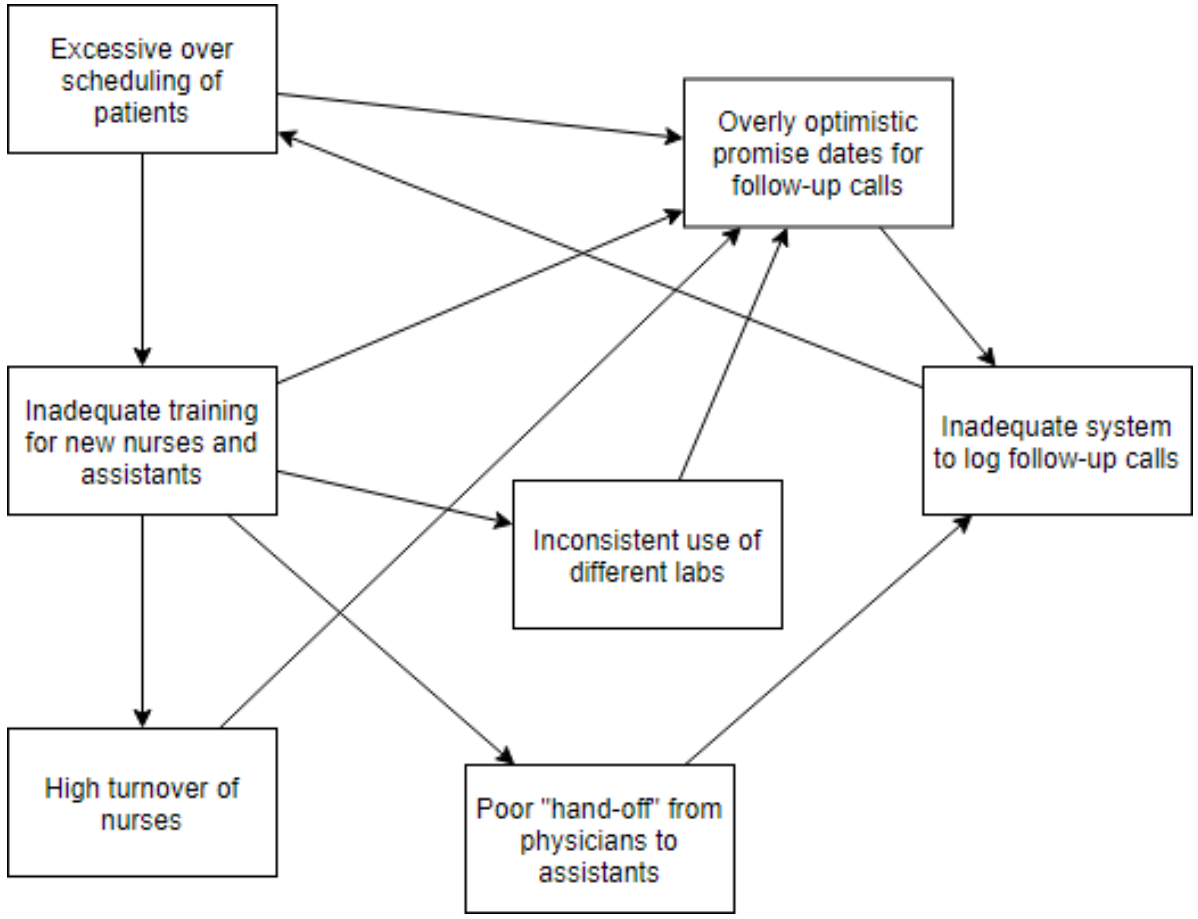
- After a brainstorming exercise
- When analysing verbal data, such as survey results
- When collecting and organising large data sets
- When developing relationships or themes among ideas
- When reducing attributes to categories that can be addressed at a higher level





Interrelationship diagram

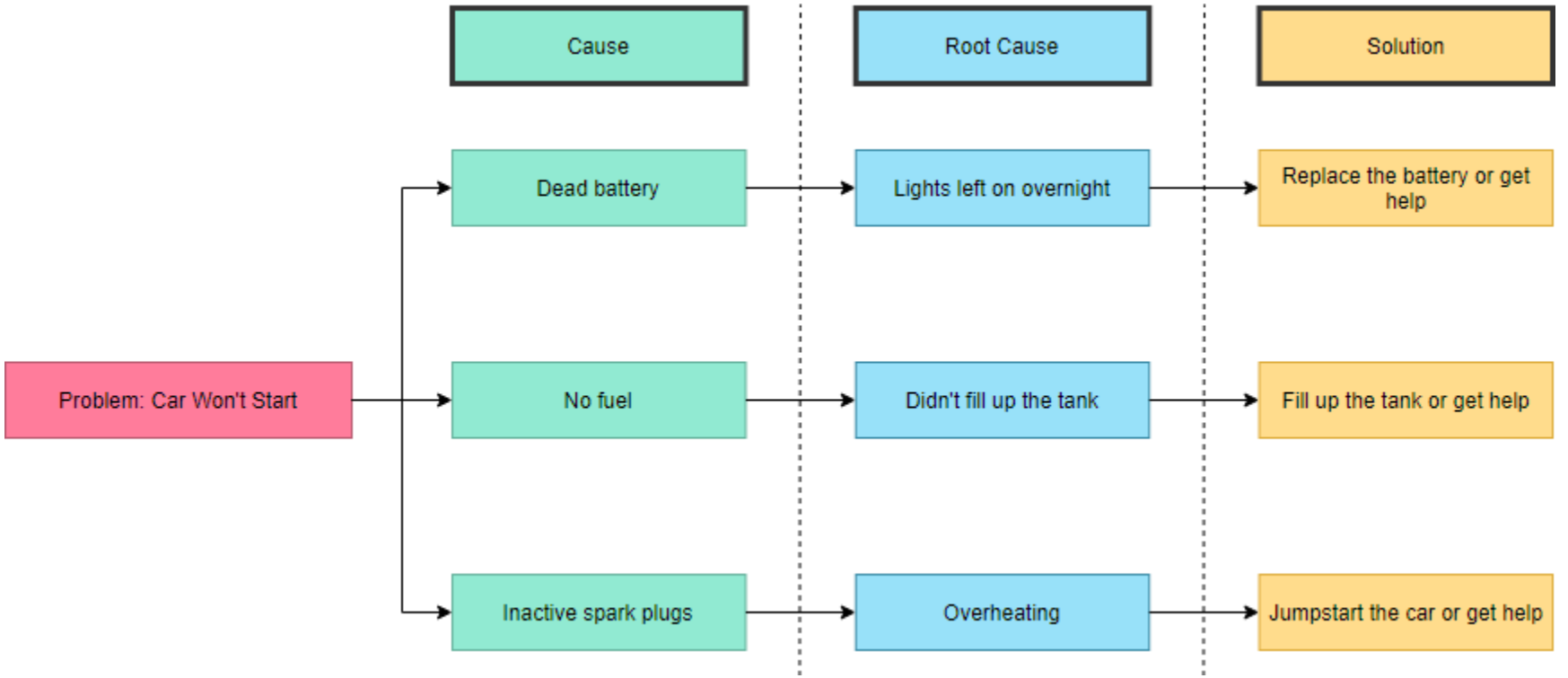
- Interrelationship Diagram is an analysis tool that helps you discover the cause and effect among critical issues.
- Interrelationship diagrams use arrows to show relationships, such as the inputs, outputs, and interconnections among processes (Dias and Saraiva, 2004).
- Arrows are used to show the direction of influences; one item may influence many other items or be influenced by many other items.



When to use an interrelationship diagram?

- When trying to understand links between ideas or cause-and-effect relationships, such as when trying to identify an area of greatest impact for improvement
- When a complex issue is being analyzed for causes
- When a complex solution is being implemented
- After generating an affinity diagram, fishbone diagram, or tree diagram to more completely explore the relations of ideas
- In root cause analysis, particularly for:
    - Understanding how different aspects of the problem are connected
    - Seeing relationships between the problem and its possible causes that can be further analyzed

Tree diagram



When to use tree diagram?

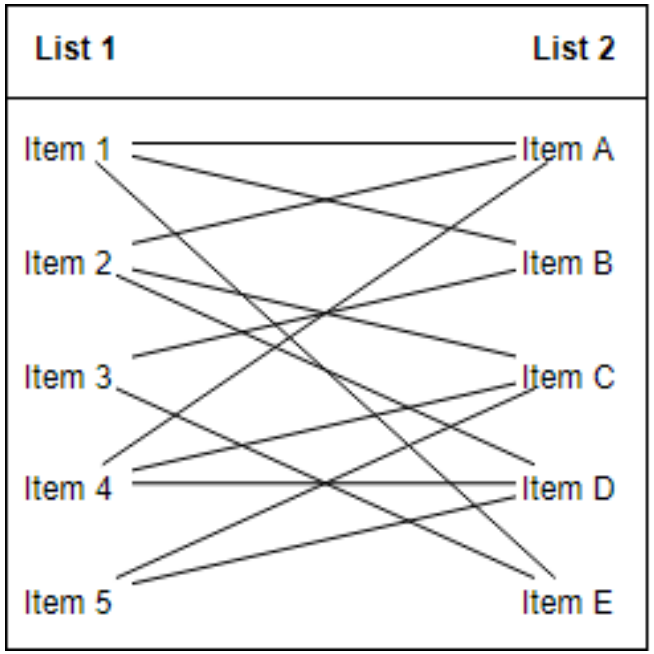
- When an issue is known or being addressed in broad generalities and must move to specific details, such as when developing logical steps to achieve an objective
- When developing actions to carry out a solution or other plan
- When analyzing processes in detail
- When probing for the root cause of a problem
- When evaluating implementation issues for several potential solutions
- After an affinity diagram or interrelationship diagram has uncovered key issues
- As a communication tool, to explain details to others

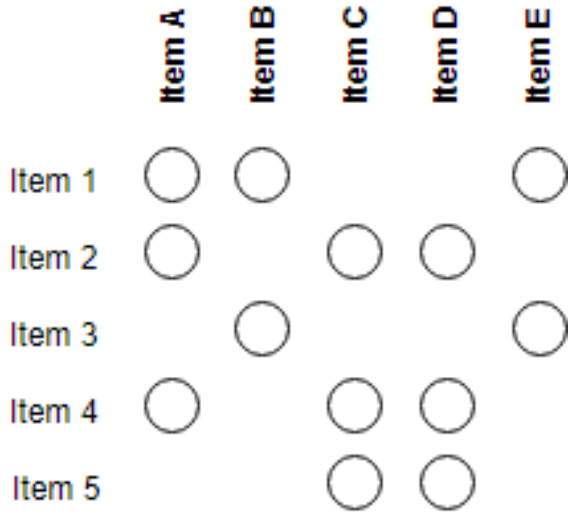




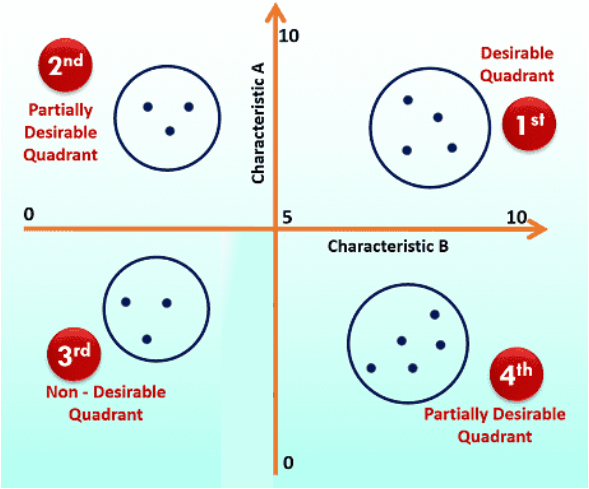
Matrix diagram

- It illustrates the critical relationships between or among two or more groups.
- It can be enhanced to show things like the strength of the relationship or the different aspects of the relationship.



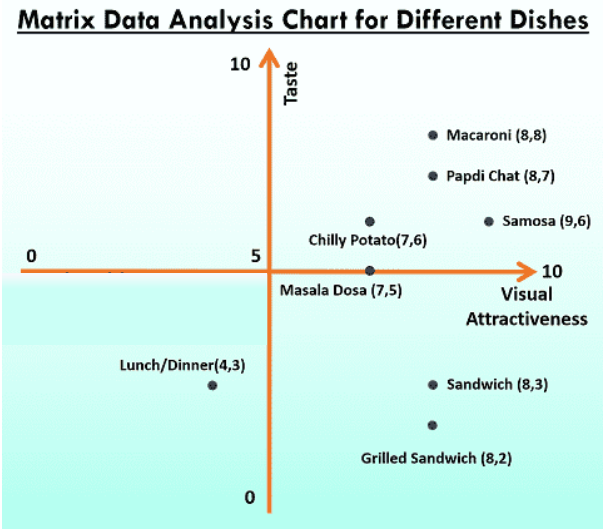


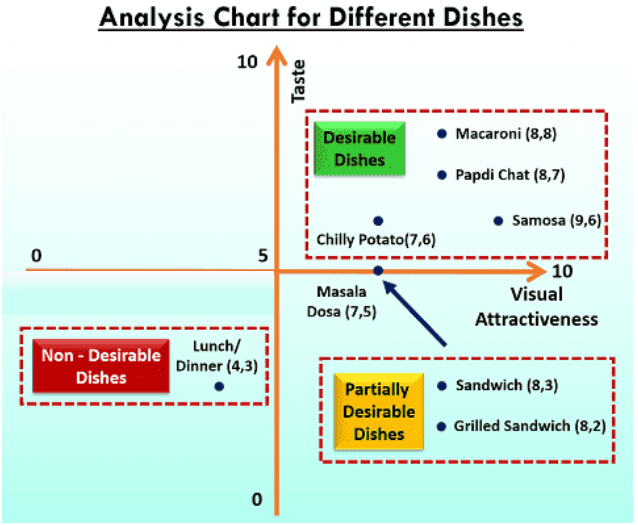
Prioritization matrices



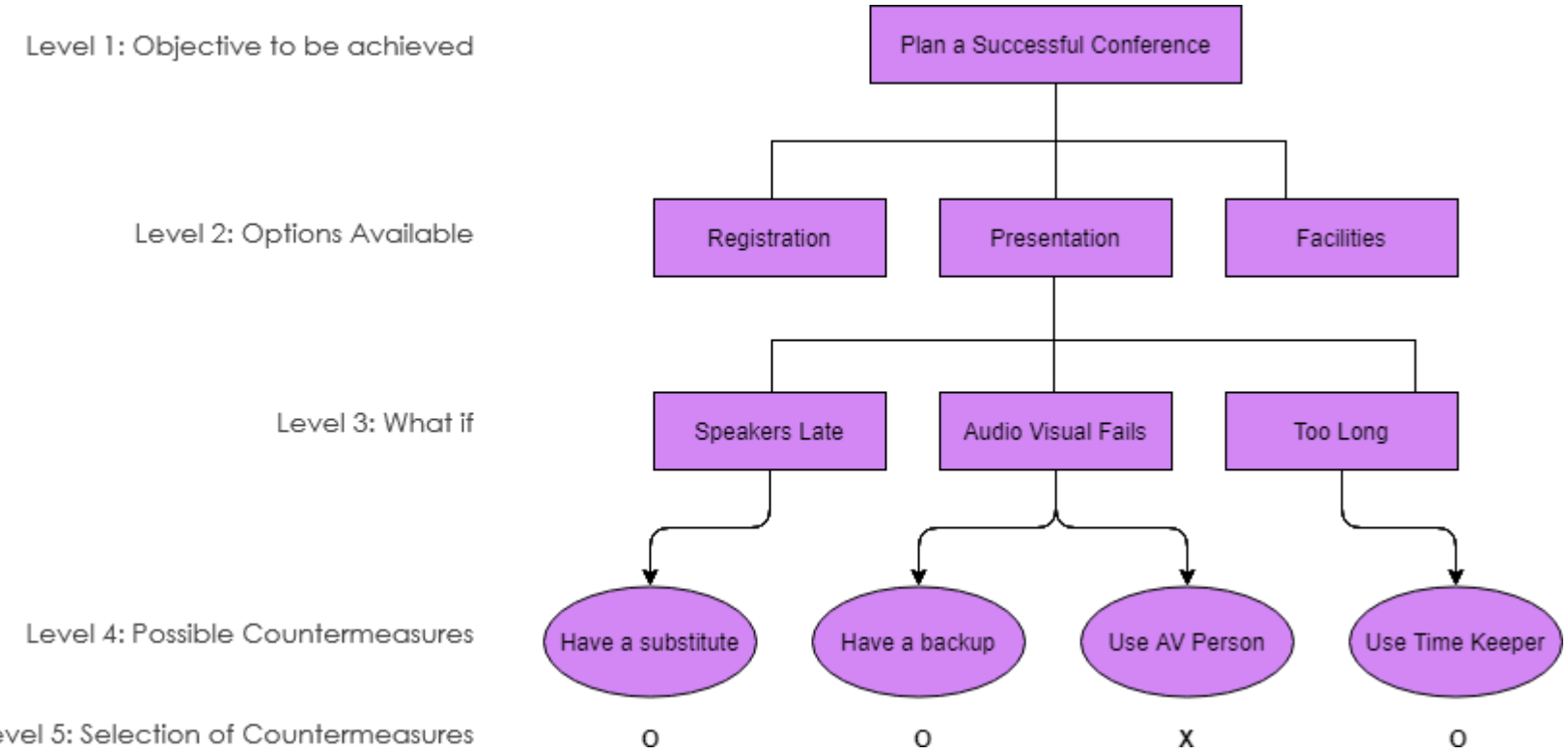
Scoring of Different Dishes

Dishes	Score for Visual Attractiveness	Score for Taste
Chilly Potato	7	6
Masala Dosa	7	5
Papdi Chat	8	7
Macaroni	8	8
Lunch/Dinner	4	3
Sandwich	8	3
Grilled Sandwich	8	2
Samosa	9	6

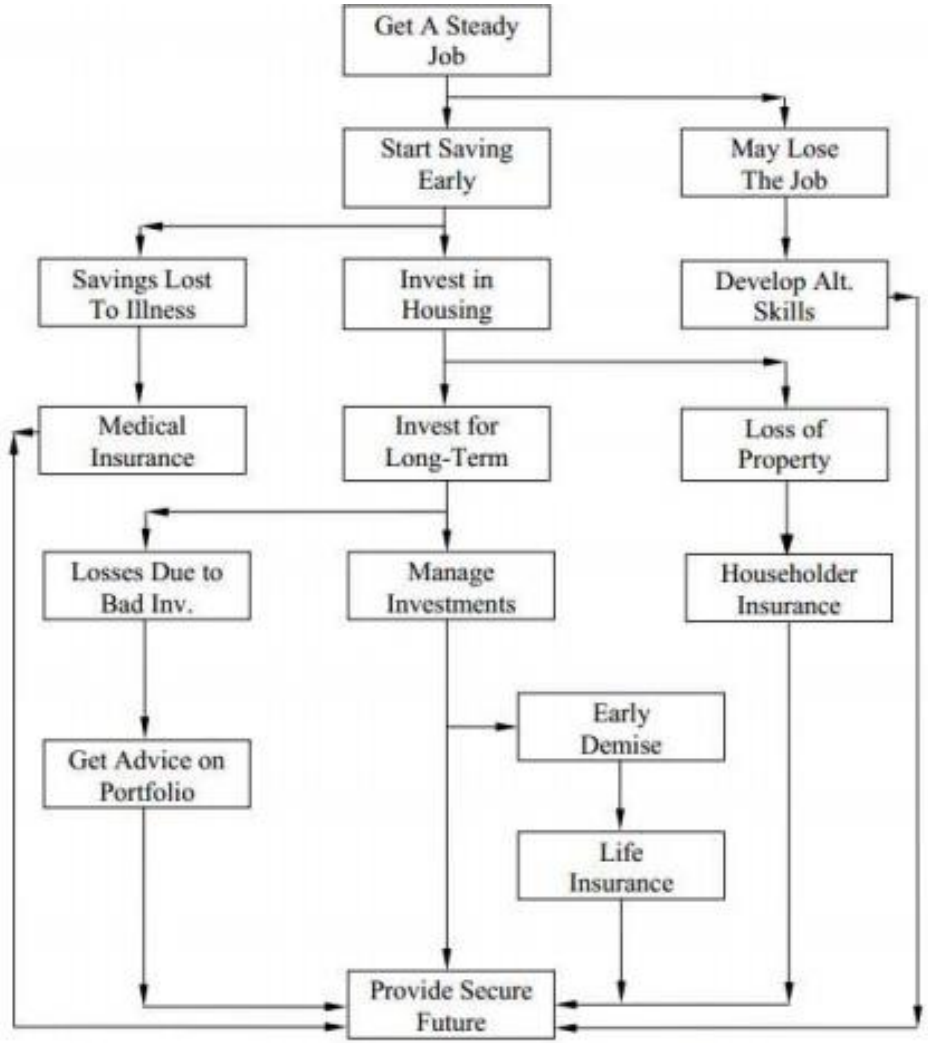




Process decision Program chart (PDPC)



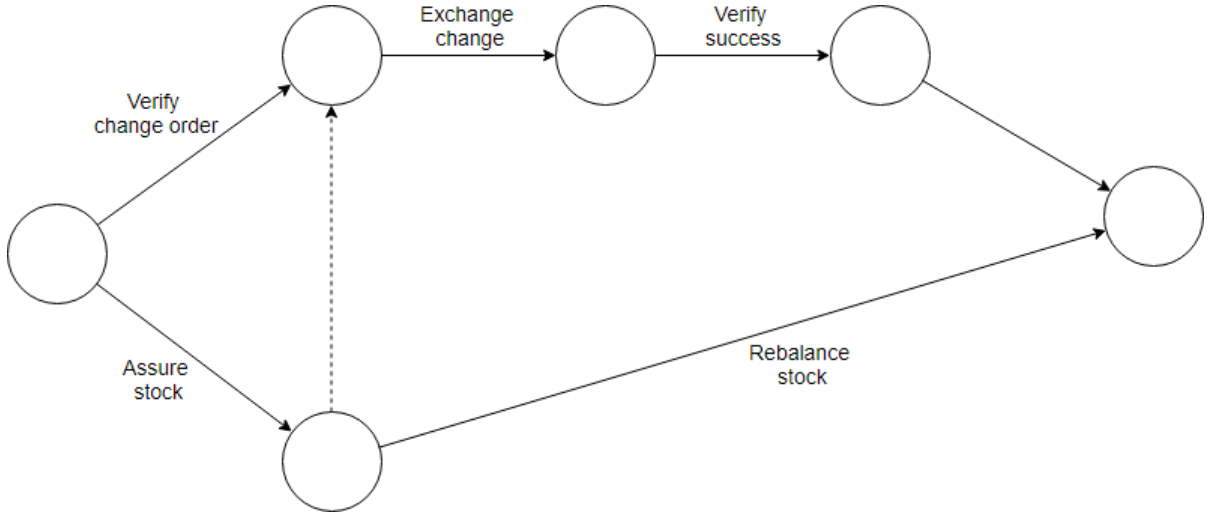
Process decision Program chart (PDPC)



An Example of PDPC - Providing Secure Future for Dependents

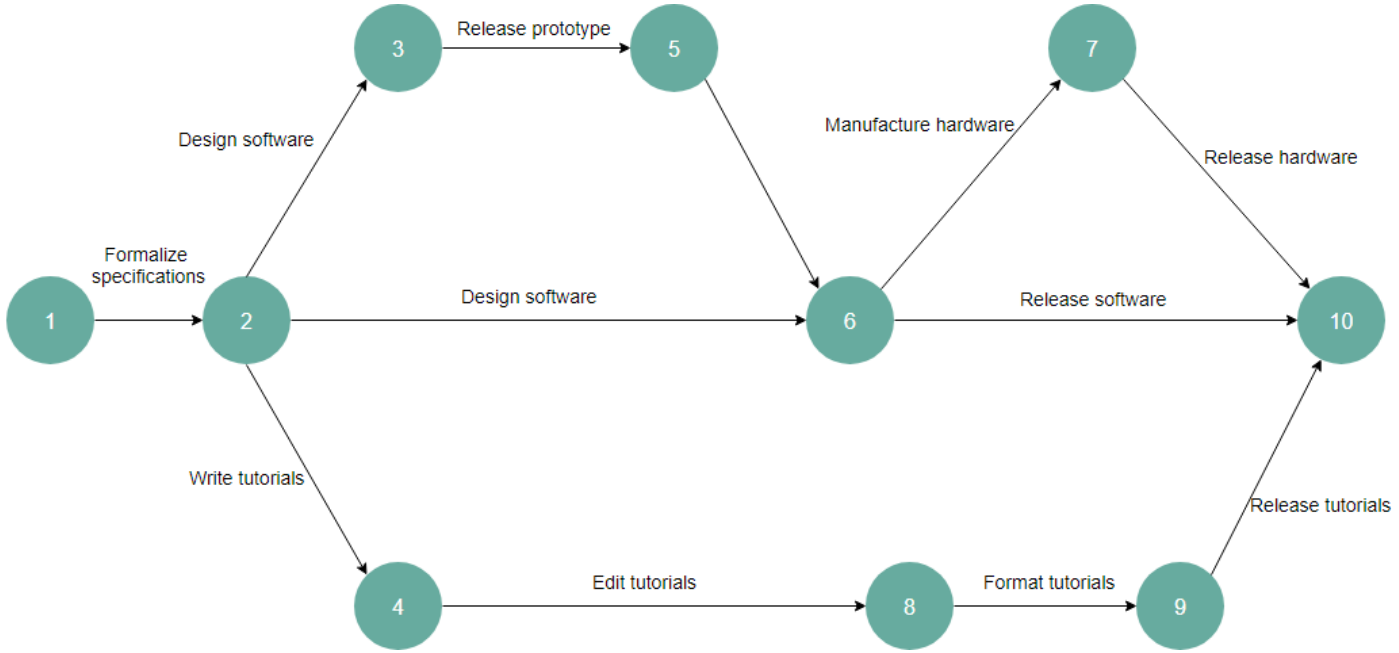
Activity Network Diagram

- How the individual tasks in a project are related, i.e. their dependencies.
- The possibility of potential problems with timings and resources.
- The possibility of potential program improvements through resource additions or program changes.
- The best schedule for the project, i.e. The critical path.
- The effect of task delays on the overall project schedule.



When to use arrow diagram?

- When scheduling and monitoring tasks within a complex project or process with interrelated tasks and resources
- When you know the steps of the project or process, their sequence, and how long each step takes
- When project schedule is critical, with serious consequences for completing the project late or significant advantage to completing the project early



Benchmarking

- **Benchmarking** is comparing the cost, cycle time, productivity, or quality of a specific process or method to another that is widely considered an industry standard or best practice.

- **Cobblers** first used the term benchmarking to measure one's feet for shoes.

- Benchmarking is mostly used to measure performance using a specific indicator, resulting in a metric of performance that is then compared to others.

Examples:

Cost per unit of measure, productivity per unit of measure, cycle time of x per unit of measure or defects per unit of measure

Benchmarking

- Benchmarking is a systematic method by which organizations can measure themselves against the best industry practices.
- It promotes superior performance by providing an organized framework through which organizations learn how the "**best in class**" do things, understand how these best practices differ from their own, and implement change to close the gap.
- The essence of benchmarking is the process of borrowing ideas and adapting them to gain a competitive advantage.
- It is a tool for continuous improvement.





Steps involved in benchmarking

1. Choose a product, service, or internal department to benchmark

2. Determine which **best-in-class** companies you should benchmark against – *which organizations you'll compare your business to*

3. Gather information on their internal performance or metrics

4. Compare the data from both organizations to identify gaps in your company's performance

5. Adopt the processes and policies in place within the best-in-class performers





Process of benchmarking

Organizations that benchmark, adapt the process to best fit their own needs and culture.

1. Decide what to benchmark.
2. Understand current performance.
3. Plan.
4. Study others.
5. Learn from the data.
6. Use the findings.





Importance of benchmarking

- Improve processes and procedures.
- Gauge the effectiveness of past performance.
- Gives a better idea of how the competition operates, which helps to identify best practices to increase performance.
- Increase efficiency and lower costs, making the business more profitable.
- Improve quality and customer satisfaction.

Types of Benchmarking

- Process benchmarking
- Financial benchmarking
- Performance benchmarking
- Product benchmarking
- Strategic benchmarking
- Functional benchmarking





Types of Benchmarking

- **Process benchmarking -** The initiating firm focuses its observation and investigation of business processes with a goal of identifying and observing the best practices from one or more benchmark firms.
- **Financial benchmarking -** Performing a financial analysis and comparing the results in an effort to assess your overall competitiveness.
- **Performance benchmarking -** Allows the initiator firm to assess their competitive position by comparing products and services with those of target firms.
- **Product benchmarking -** The process of designing new products or upgrades to current ones.
- **Strategic benchmarking -** Involves observing how others compete.
- **Functional benchmarking -** A company will focus its benchmarking on a single function in order to improve the operation of that particular function.

Benchmarking process in industries

AT & T's 12 step process

1. Determine who the clients are—who will use the information to improve their processes.
2. Advance the clients from the literacy stage to the champion stage.
3. Test the environment. Make sure the clients can and will follow through with benchmarking findings.
4. Determine urgency. Panic or disinterest indicate little chance for success.
5. Determine scope and type of benchmarking needed.
6. Select and prepare the team.
7. Overlay the benchmarking process onto the business planning process.
8. Develop the benchmarking plan.
9. Analyze the data.
10. Integrate the recommended actions.
11. Take action.
12. Continue improvement.



Xerox's 10 step process

1. Identify what is to be benchmarked.
2. Identify comparative organizations.
3. Determine data-collection method and collect data.
4. Determine current performance gap.
5. Project future performance levels.
6. Communicate benchmark findings and gain acceptance.
7. Establish functional goals.
8. Develop action plans.
9. Implement specific actions and monitor progress.
10. Recalibrate benchmarks.



THANK YOU!!!



